Homework I — Solutions

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April 13, 2026

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Problem 1

1.A)

Let an element of the Lorentz group be represented by,

$$U(\Lambda) = \exp\left(\frac{\mathrm{i}}{2}\omega_{\mu\nu}M^{\mu\nu}\right) \tag{1.1}$$

this is indirectly defined through the defining representation of the Lorentz group, which sets up the composition law, and also the inverse,

$$\Lambda^{\alpha}{}_{\beta}(\omega) \to U(\Lambda_1)U(\Lambda_2) = U(\Lambda_1\Lambda_2), \quad U^{-1}(\Lambda) = U\big(\Lambda^{\mathrm{T}}\big) \tag{1.2}$$

the Lie algebra of a group measures the non-commutative aspect of different group transformations, hence, an way in which the Lie algebra will show up is when analyzing composition of transformations, the best choice of composition of transformations to look is doing a transformation 1, after doing a transformation 2 and then undoing the transformation 1, this composition measures by how much the transformation 1 affects the way the transformation 2 should act, in other words, measures the non-commutative aspect of both transformations,

$$U(\Lambda_1)U(\Lambda_2)U^{-1}(\Lambda_1) = U(\Lambda_1\Lambda_2\Lambda_1^{\mathrm{T}}) \tag{1.3}$$

now expand on $\Lambda_2 = \mathbb{1} + \omega_2$,

$$U(\Lambda_1)U(\mathbb{1} + \omega_2)U^{-1}(\Lambda_1) = U(\Lambda_1(\mathbb{1} + \omega_2)\Lambda_1^{\mathrm{T}}) \tag{1.4}$$

$$U(\Lambda_1)\left(\mathbb{1} + \frac{\mathrm{i}}{2}\omega_{2\mu\nu}M^{\mu\nu}\right)U^{-1}(\Lambda_1) = U(\mathbb{1} + \Lambda_1\omega_2\Lambda_1^{\mathrm{T}}) \tag{1.5}$$

$$\mathbb{1} + U(\Lambda_1)\frac{\mathrm{i}}{2}\omega_{2\mu\nu}M^{\mu\nu}U^{-1}(\Lambda_1) = \mathbb{1} + \frac{\mathrm{i}}{2}\big(\Lambda_1\omega_2\Lambda_1^{\mathrm{T}}\big)_{\alpha\beta}M^{\alpha\beta} \tag{1.6}$$

$$\omega_{2\mu\nu}U(\Lambda_1)M^{\mu\nu}U^{-1}(\Lambda_1) = \Lambda_{1\alpha}{}^{\mu}\omega_{2\mu\nu}\Lambda_{1\beta}{}^{\nu}M^{\alpha\beta} \tag{1.7}$$

$$U(\Lambda_1)M^{\mu\nu}U^{-1}(\Lambda_1) = \Lambda_{1\alpha}{}^{\mu}\Lambda_{1\beta}{}^{\nu}M^{\alpha\beta} \tag{1.8}$$

in this last step we factored out ω_2 by using that both sides are antisymmetric in $\mu\nu$. This equation already is useful by itself, as it shows how the generators themselves transforms under a group action, but, as we're interested into the algebra of the group, we still need to expand around Λ_1,

$$U(\mathbb{1} + \omega_1)M^{\mu\nu}U^{-1}(\mathbb{1} + \omega_1) = (g_{\alpha}{}^{\mu} + \omega_{1\alpha}{}^{\mu})\big(g_{\beta}{}^{\nu} + \omega_{1\beta}{}^{\nu}\big)M^{\alpha\beta} \tag{1.9}$$

$$\left(\mathbb{1} + \frac{\mathrm{i}}{2}\omega_{1\alpha\beta}M^{\alpha\beta}\right)M^{\mu\nu}\left(\mathbb{1} - \frac{\mathrm{i}}{2}\omega_{1\alpha\beta}M^{\alpha\beta}\right) = M^{\mu\nu} + M^{\mu\beta}\omega_{1\beta}{}^{\nu} + M^{\alpha\nu}\omega_{1\alpha}{}^{\mu} \tag{1.10}$$

$$M^{\mu\nu} + \frac{\mathrm{i}}{2}\omega_{1\alpha\beta}M^{\alpha\beta}M^{\mu\nu} - \frac{\mathrm{i}}{2}\omega_{1\alpha\beta}M^{\mu\nu}M^{\alpha\beta} = M^{\mu\nu} + M^{\mu\beta}\omega_{1\beta}{}^{\nu} + M^{\alpha\nu}\omega_{1\alpha}{}^{\mu} \tag{1.11}$$

$$\frac{\mathrm{i}}{2}\omega_{1\alpha\beta}\big[M^{\alpha\beta}, M^{\mu\nu}\big] = \omega_{1\alpha\beta}\big(-g^{\alpha\nu}M^{\mu\beta} + M^{\alpha\nu}g^{\beta\mu}\big) \tag{1.12}$$

to factorize ω_1 we need to ensure both sides are antisymmetric, this can be simply done by antisymmetrizing the right hand side,

$$\frac{\mathrm{i}}{2}\omega_{1\alpha\beta}\big[M^{\alpha\beta}, M^{\mu\nu}\big] = \frac{1}{2}\omega_{1\alpha\beta}\big(-g^{\alpha\nu}M^{\mu\beta} + M^{\alpha\nu}g^{\beta\mu} + g^{\beta\nu}M^{\mu\alpha} - M^{\beta\nu}g^{\alpha\mu}\big) \tag{1.13}$$

$$\mathrm{i}\big[M^{\alpha\beta}, M^{\mu\nu}\big] = -g^{\alpha\nu}M^{\mu\beta} + M^{\alpha\nu}g^{\beta\mu} + g^{\beta\nu}M^{\mu\alpha} - M^{\beta\nu}g^{\alpha\mu} \tag{1.14}$$

The sign on the right hand side is dependent of metric convention, here the answer is for the $\begin{pmatrix} - & + & + & + \end{pmatrix}$ metric, the opposite sign metric has as answer an additional sign in the right hand side.

1.B)

In the last item we obtained the Lie algebra $\mathfrak{so}^+(1,3)$, or as we argued $\mathfrak{sl}(2,\mathbb{C})$. The usual way of constructing representations of Lie algebras is to do by the highest weight method, that is, we set up all the Casimirs of the algebra and choose a maximal set of mutually commuting algebra elements, then we construct a vector space in which these chosen operators are diagonal, and lastly with the remaining algebra elements construct ladder operators which acts on the vector space by lowering and increasing the eigenvalues of the chosen mutually commuting operators. As a remainder — and as will be relevant — the main example of this process which is physically relevant is the computation of the representations of the group $SO(3) \cong SU(2)/\mathbb{Z}_2$, which have the algebra $\mathfrak{so}(3) \cong \mathfrak{su}(2)$,

$$\left[J^i, J^j\right] = \mathrm{i}\epsilon^{ij}{}_k J^k, \quad i,j,k = 1,2,3 \tag{1.15}$$

the number of Casimirs of an algebra is not a trivial thing to obtain, for special cases, as semi-simple algebras, the number of independent Casimirs is equal to the rank of the algebra. Luckily, there is a complete classification of all the simple Lie algebras and their ranks, the $\mathfrak{su}(2)$ algebra is simple of rank one, then there is just one independent Casimir, $C = \delta_{ij}J^iJ^j$. Usually J^3 is chosen as part of the maximal set of mutually commuting algebra elements — in this case the *set* is just one element —. Representations are then indexed by two numbers,

$$C\left|j,m\right\rangle = j(j+1)\left|k,m\right\rangle, \quad J^3\left|j,m\right\rangle = m\left|j,m\right\rangle, \quad j \in \frac{1}{2}\mathbb{N}, \quad m = -j, -j+1, \cdots, j-1, j \tag{1.16}$$

The same process will happen for the Lorentz algebra, but now the rank of the algebra is two, it can be verified that the two Casimirs are,

$$C_1 = \frac{1}{2}M^{\mu\nu}M_{\mu\nu}, \quad C_2 = \frac{1}{4}\epsilon_{\mu\nu\alpha\beta}M^{\mu\nu}M^{\alpha\beta} \tag{1.17}$$

sadly, is not that enlightening to workout the representations in a covariant manner, it's easier to decompose $M^{\mu\nu}$ into it's spatial parts,

$$J_i = \frac{1}{2}\epsilon_{ijk}M^{jk}, \quad K^i = M^{i0} \tag{1.18}$$

now we have to compute the algebra in terms of these new generators, first the $J-J$ commutation,

$$\begin{aligned}
[J_i, J_j] &= \frac{1}{4}\epsilon_{iab}\epsilon_{jlm}\left[M^{ab}, M^{lm}\right] && (1.19)\\
&= \frac{\mathrm{i}}{4}\epsilon_{iab}\epsilon_{jlm}\left(g^{al}M^{bm} - g^{bl}M^{am} + g^{bm}M^{al} - g^{am}M^{bl}\right) && (1.20)\\
&= \frac{\mathrm{i}}{2}\epsilon_{iab}\epsilon_{jlm}\left(g^{al}M^{bm} + g^{bm}M^{al}\right) && (1.21)\\
&= \frac{\mathrm{i}}{2}\epsilon_{iab}\epsilon_{jlm}\left(g^{al}M^{bm} - g^{am}M^{bl}\right) && (1.22)\\
&= \mathrm{i}\epsilon_{iab}\epsilon_{jlm}g^{al}M^{bm} = \mathrm{i}\epsilon_i{}^l{}_b\epsilon_{jlm}M^{bm} && (1.23)\\
&= \mathrm{i}\epsilon^l{}_{bi}\epsilon_{lmj}M^{bm} = \mathrm{i}\left(g_{bm}g_{ij} - g_{bj}g_{im}\right)M^{bm}, \quad \text{antisymmetrize } (b \leftrightarrow m) && (1.24)
\end{aligned}$$

$$= -\frac{\mathrm{i}}{2}\big(g_{bj}g_{mi} - g_{bi}g_{mj}\big)M^{bm} = \frac{\mathrm{i}}{2}\epsilon_{lbm}\epsilon^{l}{}_{ij}M^{bm} \tag{1.25}$$
$$= \mathrm{i}\epsilon_{ij}{}^{l}J_l \tag{1.26}$$

now the $J - K$,

$$\big[J_i, K^j\big] = \frac{1}{2}\epsilon_{iab}\big[M^{ab}, M^{j0}\big] \tag{1.27}$$
$$= \frac{\mathrm{i}}{2}\epsilon_{iab}\big(g^{aj}M^{b0} - g^{bj}M^{a0} + g^{b0}M^{aj} - g^{a0}M^{bj}\big) \tag{1.28}$$
$$= \frac{\mathrm{i}}{2}\epsilon_{iab}2g^{aj}M^{b0} = \mathrm{i}\epsilon_{i}{}^{j}{}_{b}M^{b0} \tag{1.29}$$
$$= \mathrm{i}\epsilon_{i}{}^{j}{}_{k}K^k \tag{1.30}$$

lastly $K - K$,

$$\big[K^i, K^j\big] = \big[M^{i0}, M^{j0}\big] \tag{1.31}$$
$$= \mathrm{i}\big(g^{ij}M^{00} - g^{i0}M^{0j} + g^{00}M^{ij} - g^{0j}M^{i}{}_{0}\big) \tag{1.32}$$
$$= -\mathrm{i}M^{ij} = -\mathrm{i}g^{i}{}_{a}g^{j}{}_{b}M^{ab}, \quad \text{antisymmetrize } (a \leftrightarrow b) \tag{1.33}$$
$$= -\frac{\mathrm{i}}{2}\big(g^{i}{}_{a}g^{j}{}_{b} - g^{i}{}_{b}g^{j}{}_{a}\big)M^{ab} \tag{1.34}$$
$$= -\frac{\mathrm{i}}{2}\epsilon^{lij}\epsilon_{lab}M^{ab} \tag{1.35}$$
$$= -\mathrm{i}\epsilon^{ijk}J_k \tag{1.36}$$

Just for clarity sake, let's rewrite this new representation of the full $\mathfrak{sl}(2,\mathbb{C})$ algebra,

$$[J_i, J_j] = \mathrm{i}\epsilon_{ij}{}^{k}J_k \tag{1.37}$$
$$\big[J_i, K^j\big] = \mathrm{i}\epsilon_{i}{}^{j}{}_{k}K^k \tag{1.38}$$
$$\big[K^i, K^j\big] = -\mathrm{i}\epsilon^{ijk}J_k \tag{1.39}$$

we just have been careful with the index placement, but, for our choice of metric $g_{ij} = \delta_{ij}$, thus, $J^i = J_i, K^i = K_i$.

The usual treatment is to define now two new combination of generators,

$$N_i = \frac{1}{2}(J_i - \mathrm{i}K_i), \quad N_i^\dagger = \frac{1}{2}(J_i + \mathrm{i}K_i) \tag{1.40}$$

this is not entirely justifiable, at the most fundamental level a Lie algebra is a vector space over the **real** numbers, a multiplication by some imaginary quantity is not well defined, as it even does not preserve the structure of the Lie bracket. The working procedure is to consider a complexification of a real Lie algebra, $\mathfrak{g} \hookrightarrow \mathfrak{g}_\mathbb{C}$ where the complexified Lie algebra $\mathfrak{g}_\mathbb{C}$ is understood as a real Lie algebra. This is done by considering the generators $T^I \in \mathfrak{g}$ linearly independent from $\mathrm{i}T^I$. Let's illustrate this process by working what is the complexification of the $\mathfrak{su}(2)$ algebra, we start from the algebra, eq. (1.15), and define the complexified algebra $\mathfrak{su}(2)_\mathbb{C}$ to be generated by $J^i, \mathrm{i}J^i$, to stress that we are considering them to be linearly independent, we are going to rename $\mathrm{i}J^i$ to K^i. Now we compute the algebra from eq. (1.15) with these new set of generators, first between J^i and $\mathrm{i}J^j = K^j$

$$\big[J^i, K^j\big] = \big[J^i, \mathrm{i}J^j\big] = \mathrm{i}\big[J^i, J^j\big] = \mathrm{i}^2\epsilon^{ij}{}_{k}J^k = \mathrm{i}\epsilon^{ij}{}_{k}\mathrm{i}J^k = \mathrm{i}\epsilon^{ij}{}_{k}K^k \tag{1.41}$$

now between $\mathrm{i}J^i = K^i$ and $\mathrm{i}J^j = K^j$,

$$\big[K^i, K^j\big] = \big[\mathrm{i}J^i, \mathrm{i}J^j\big] = \mathrm{i}^2\big[J^i, J^j\big] = \mathrm{i}^2\mathrm{i}\epsilon^{ij}{}_{k}J^k = -\mathrm{i}\epsilon^{ij}{}_{k}J^k \tag{1.42}$$

for clarity, the algebra we found for $\mathfrak{su}(2)_{\mathbb{C}}$ in the basis $J^i, \mathrm{i}J^i = K^i$ is,

$$\left[J^i, J^j\right] = \mathrm{i}\epsilon^{ij}{}_k J^k \tag{1.43}$$
$$\left[J^i, K^j\right] = \mathrm{i}\epsilon^{ij}{}_k K^k \tag{1.44}$$
$$\left[K^i, K^j\right] = -\mathrm{i}\epsilon^{ij}{}_k J_k \tag{1.45}$$

exactly the $\mathfrak{sl}(2,\mathbb{C})$ algebra! In other words $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2,\mathbb{C})$.

After this detour, we can go back to introducing eq. (1.40), which now we're aware that implies we're complexifying $\mathfrak{sl}(2,\mathbb{C})$, the new basis of generators in the complexified algebra $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{C}}$ are $J^i, K^i, \mathrm{i}J^i, \mathrm{i}K^i$, which can be exchanged by $N_i, \mathrm{i}N_i, N_i^\dagger, \mathrm{i}N_i^\dagger$. It suffices to work out the $N - N^\dagger$ algebra, the others will follow, first $N - N$,

$$[N_i, N_j] = \frac{1}{4}[J_i - \mathrm{i}K_i, J_j - \mathrm{i}K_j] \tag{1.46}$$
$$= \frac{\mathrm{i}}{4}\epsilon_{ijk}J^k + \frac{1}{4}\epsilon_{ijk}K^k + \frac{1}{4}\epsilon_{ijk}K^k + \frac{\mathrm{i}}{4}\epsilon_{ijk}J^k \tag{1.47}$$
$$= \mathrm{i}\epsilon_{ijk}\frac{1}{2}\left(J^k - \mathrm{i}K^k\right) \tag{1.48}$$
$$= \mathrm{i}\epsilon_{ijk}N^k \tag{1.49}$$

then $N^\dagger - N^\dagger$,

$$\left[N_i^\dagger, N_j^\dagger\right] = \frac{1}{4}[J_i + \mathrm{i}K_i, J_j + \mathrm{i}K_j] \tag{1.50}$$
$$= \frac{\mathrm{i}}{4}\epsilon_{ijk}J^k - \frac{1}{4}\epsilon_{ijk}K^k - \frac{1}{4}\epsilon_{ijk}K^k + \frac{\mathrm{i}}{4}\epsilon_{ijk}J^k \tag{1.51}$$
$$= \mathrm{i}\epsilon_{ijk}\frac{1}{2}\left(J^k + \mathrm{i}K^k\right) \tag{1.52}$$
$$= \mathrm{i}\epsilon_{ijk}N^{\dagger k} \tag{1.53}$$

and lastly $N - N^\dagger$,

$$\left[N_i, N_j^\dagger\right] = \frac{1}{4}[J_i - \mathrm{i}K_i, J_j + \mathrm{i}K_j] \tag{1.54}$$
$$= \frac{\mathrm{i}}{4}\epsilon_{ijk}j^k - \frac{1}{4}\epsilon_{ijk}K^k + \frac{1}{4}\epsilon_{ijk}K^k - \frac{\mathrm{i}}{4}\epsilon_{ijk}J^k \tag{1.55}$$
$$= 0 \tag{1.56}$$

together with the remaining commutations of $\mathrm{i}N, \mathrm{i}N^\dagger$. Thus, to summarize, the algebra of $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{C}}$ is,

$$[N_i, N_j] = \mathrm{i}\epsilon_{ij}{}^k N_k \tag{1.57}$$
$$\left[N_i^\dagger, N_j^\dagger\right] = \mathrm{i}\epsilon_{ij}{}^k N_k^\dagger, \qquad \text{Plus commutators involving } \mathrm{i}N, \mathrm{i}N^\dagger \tag{1.58}$$
$$\left[N_i, N_j^\dagger\right] = 0 \tag{1.59}$$

but notice, the algebra generated by N alone is $\mathfrak{su}(2)$, as is the one generated by $N^\dagger$ alone, by the commutation relations the algebra from N is decoupled from the algebra of $N^\dagger$, thus, to go from the algebra generated by N to the algebra generated by $N, \mathrm{i}N$ is simply the complexification of the algebra of N. Hence, the algebra generated by $N, \mathrm{i}N$ is $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2,\mathbb{C})$, the same is valid for the pair $N^\dagger, \mathrm{i}N^\dagger$. As the algebras between N and $N^\dagger$ are independent, so are the algebras

between $N, \mathrm{i}N$ and $N^\dagger, \mathrm{i}N^\dagger$, what implies that $\mathfrak{sl}(2,\mathbb{C})_\mathbb{C}$, which is the algebra generated by $N, \mathrm{i}N, N^\dagger, \mathrm{i}N^\dagger$ is just the sum of the two independent algebras, in other words,

$$\mathfrak{sl}(2,\mathbb{C})_\mathbb{C} \cong \mathfrak{su}(2)_\mathbb{C} \oplus \mathfrak{su}(2)_\mathbb{C} \tag{1.60}$$
$$\mathfrak{sl}(2,\mathbb{C})_\mathbb{C} \cong \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}) \tag{1.61}$$

The first form of this result is of the most significance for us,

$$\mathfrak{sl}(2,\mathbb{C})_\mathbb{C} \cong \mathfrak{su}(2)_\mathbb{C} \oplus \mathfrak{su}(2)_\mathbb{C} \cong (\mathfrak{su}(2) \oplus \mathfrak{su}(2))_\mathbb{C} \tag{1.62}$$

Thus, the real representations of $\mathfrak{sl}(2,\mathbb{C})_\mathbb{C}$ are into a one-to-one map with the real representations of $(\mathfrak{su}(2) \oplus \mathfrak{su}(2))_\mathbb{C}$, what is equivalent to say that the representations of $\mathfrak{sl}(2,\mathbb{C})$ are into a one-to-one with representations of $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$. Of course, these two $\mathfrak{su}(2)$ are the ones generated by $N, N^\dagger$.

Is important to stress that eq. (1.62) **does not** imply that $\mathfrak{sl}(2,\mathbb{C})$ is isomorphic to $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, the process of complexification does not preserve crucial topological structure, as compactness. $\mathfrak{sl}(2,\mathbb{C})$ is a non-compact algebra, while $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ is compact. This is sufficient to conclude $\mathfrak{sl}(2,\mathbb{C}) \ncong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$.

This whole convoluted argument is just to justify why is ok — in this case — to multiply the generators by complex numbers and still obtain a reasonable answer. To finish to find the representation of the Lorentz algebra, we continue as always in the usual procedure of finding the representations of $\mathfrak{su}(2)$ which we already described, there are two Casimirs,

$$C_L = \delta^{ij} N_i N_j, \quad C_R = \delta^{ij} N_i^\dagger N_j^\dagger \tag{1.63}$$

choosing also as a mutually commuting set of algebra elements $N_3, N_3^\dagger$, then the representations are given by four numbers,

$$C_L \left|j_L, m_L; j_R, m_R\right\rangle = j_L(j_L+1) \left|j_L, m_L; j_R, m_R\right\rangle, \; C_R \left|j_L, m_L; j_R, m_R\right\rangle = j_R(j_R+1) \left|j_L, m_L; j_R, m_R\right\rangle \tag{1.64}$$

$$N_3 \left|j_L, m_L; j_R, m_R\right\rangle = m_L \left|j_L, m_L; j_R, m_R\right\rangle, \; N_3^\dagger \left|j_L, m_L; j_R, m_R\right\rangle = m_R \left|j_L, m_L; j_R, m_R\right\rangle \tag{1.65}$$

$$j_{L/R} \in \frac{1}{2}\mathbb{N}, \quad m_{L/R} = -j_{L/R}, -j_{L/R}+1, \cdots, j_{L/R}-1, j_{L/R} \tag{1.66}$$

the representations are indexed by $j_{L/R}$, generally are written as,

$$(j_L, j_R) \tag{1.67}$$

The simplest possible representation is the scalar one, $j_L = j_R = 0$, this fixes,

$$(0,0) \to N_i = N_i^\dagger = 0 \Rightarrow J^i = K^i = 0 \Rightarrow M^{\mu\nu} = 0 \Rightarrow U(\Lambda) = 1 \tag{1.68}$$

$U(\Lambda) = 1$ is what we should expect from a scalar representation, as a scalar do not transform under Lorentz. Now, the simplest non trivial representation we can obtain is given by either $j_L = \frac{1}{2}, j_R = 0$ or $j_R = \frac{1}{2}, j_L = 0$ they are called the Left-handed Weyl spinorial representation and the Right-handed Weyl spinorial representation respectively. As example let's pickup the Left-handed one, we know that the $j = \frac{1}{2}$ representation of $\mathfrak{su}(2)$ is given by the Pauli matrices, then,

$$\left(\frac{1}{2}, 0\right) \to N_i = \frac{1}{2}\sigma_i, \; N_i^\dagger = 0 \Rightarrow J^i = \frac{1}{2}\sigma^i, \; K^i = \frac{\mathrm{i}}{2}\sigma^i \Rightarrow M^{ij} = \frac{1}{2}\epsilon^{ij}{}_k \sigma^k, \; M^{i0} = \frac{\mathrm{i}}{2}\sigma^i \tag{1.69}$$

the Right-handed one has a similar form. Now, a interesting representation is the vector one,

$$\left(\frac{1}{2},\frac{1}{2}\right) = \left(\frac{1}{2},0\right) \otimes \left(0,\frac{1}{2}\right) \tag{1.70}$$

to see why this should be a vector representation we need to remember some results about sum of angular momentum, when going from the $N^2, N^{\dagger 2}$ basis to the $\left(N+N^\dagger\right)^2$ basis, we exchange the $j_{L/R}$ for a total $j = |j_L - j_R|, \cdots, j_L + j_R$. Notice that $j(j+1)$ is the eigenvalue of $(N+N^\dagger)^2$, but,

$$(N+N^\dagger)^2 = J^2 \tag{1.71}$$

hence the j is exactly the angular momentum of the representation. For what we called the vector representation,

$$\left(\frac{1}{2},\frac{1}{2}\right) = \mathbf{0} \oplus \mathbf{1} \tag{1.72}$$

it carries only a scalar, under rotations, and a vector, under rotations, hence the name vector representation.

Problem 2

The remaining of the problems, this one included, are about applications of the Noether's theorem to some regular cases. Here the electromagnetic energy-momentum tensor is derived, after the equations of motion are obtained.

2.A)

For obtaining the equations of motion from the action, the usual procedure is to vary the action with respect to an infinitesimal (first order) change in the fields — this whole deal of infinitesimal changes in functionals can be formalized by a functional derivative —. We start with $S[A+\delta A]-S[A]$, then expand $F^{\mu\nu}[A+\delta A]$ in Taylor around A, and lastly integrate by parts,

$$S[A+\delta A]-S[A]=-\frac{1}{4}\int \mathrm{d}^4x\,(F^{\mu\nu}(A+\delta A)F_{\mu\nu}(A+\delta A)-F^{\mu\nu}(A)F_{\mu\nu}(A)) \tag{2.1}$$

$$S[A+\delta A]-S[A]=-\frac{1}{4}\int \mathrm{d}^4x\left(F^{\mu\nu}(A)F_{\mu\nu}(A)+2F^{\mu\nu}(A)\frac{\partial F_{\mu\nu}(A)}{\partial\partial_\alpha A_\beta}\partial_\alpha\delta A_\beta-F^{\mu\nu}(A)F_{\mu\nu}(A)\right) \tag{2.2}$$

$$S[A+\delta A]-S[A]=-\frac{1}{2}\int \mathrm{d}^4x\,F^{\mu\nu}(A)\big(g_\mu{}^\alpha g_\nu{}^\beta-g_\nu{}^\alpha g_\mu{}^\beta\big)\partial_\alpha\delta A_\beta \tag{2.3}$$

$$S[A+\delta A]-S[A]=-\frac{1}{2}\int \mathrm{d}^4x\,\big(F^{\alpha\beta}(A)-F^{\beta\alpha}(A)\big)\partial_\alpha\delta A_\beta \tag{2.4}$$

$$S[A+\delta A]-S[A]=-\int \mathrm{d}^4x\,F^{\alpha\beta}(A)\partial_\alpha\delta A_\beta \tag{2.5}$$

$$S[A+\delta A]-S[A]=-\int \mathrm{d}^4x\,\big\{\partial_\alpha\big(F^{\alpha\beta}(A)\delta A_\beta\big)-\partial_\alpha F^{\alpha\beta}(A)\delta A_\beta\big\} \tag{2.6}$$

$$S[A+\delta A]-S[A]=\int \mathrm{d}^4x\,\big(\partial_\alpha F^{\alpha\beta}(A)\big)\delta A_\beta-\int \mathrm{d}^4x\,\partial_\alpha\big(F^{\alpha\beta}(A)\delta A_\beta\big) \tag{2.7}$$

the last term is an integral of a total derivative, also called a boundary term because, by Stokes theorem, a integral over space-time of a total derivative becomes an integral over the space-time boundary. Usually the boundary term is taken to be zero by saying that the space-time has no boundary or by considering it as a constraint on the values of the fields in the asymptotical infinity regions. Either way, neglecting the boundary, for a physical configuration, the action should be an extremum, thus, the first order infinitesimal variation should vanish,

$$0=\int \mathrm{d}^4x\,\big(\partial_\alpha F^{\alpha\beta}(A)\big)\delta A_\beta \tag{2.8}$$

as this should be true for all possible variations — at lest the variations which keeps the initial value data the same, as this selects a specific solution —, we conclude that

$$0=\partial_\alpha F^{\alpha\beta} \tag{2.9}$$

is the equation of motion.

2.B)

In this item, and also in the others where is requested for a Noether's current computation, we will use the formula for the Noether's current derived in section E, specially eq. (E.21),

which when applied to our case gives,

$$T^{\mu}{}_{\nu} = \frac{\partial \mathcal{L}}{\partial \partial_\mu A_\alpha(x)} \frac{\mathrm{d}A'_\alpha(x)}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} + \mathcal{L}\frac{\mathrm{d}x'^\mu}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} \tag{2.10}$$

the missing information is the first order infinitesimal variation of the translation transformation,

$$\begin{cases} x & \to x'^\mu = x^\mu + a^\mu \\ A_\alpha(x) & \to A'{}_\alpha(x') = A_\alpha(x) \end{cases} \Rightarrow \frac{\mathrm{d}x'^\mu}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} = g^\mu{}_\nu, \quad \frac{\mathrm{d}A'_\alpha(x)}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} = -\partial_\nu A_\alpha(x) \tag{2.11}$$

it's straightforward to substitute it,

$$T^{\mu}{}_{\nu} = \frac{\partial \mathcal{L}}{\partial \partial_\mu A_\alpha(x)} \frac{\mathrm{d}A'_\alpha(x)}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} + \mathcal{L}\frac{\mathrm{d}x'^\mu}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} \tag{2.12}$$

$$T^{\mu}{}_{\nu} = -\frac{\partial \mathcal{L}}{\partial \partial_\mu A_\alpha(x)}\partial_\nu A_\alpha(x) + \mathcal{L}g^\mu{}_\nu \tag{2.13}$$

$$T^{\mu}{}_{\nu} = \frac{1}{2}F^{\rho\sigma}\frac{\partial F_{\rho\sigma}}{\partial \partial_\mu A_\alpha(x)}\partial_\nu A_\alpha(x) - \frac{1}{4}F_{\rho\sigma}F^{\rho\sigma}g^\mu{}_\nu \tag{2.14}$$

$$T^{\mu}{}_{\nu} = \frac{1}{2}F^{\rho\sigma}\big(g_\rho{}^\mu g_\sigma{}^\alpha - g_\rho{}^\alpha g_\sigma{}^\mu\big)\partial_\nu A_\alpha(x) - \frac{1}{4}F_{\rho\sigma}F^{\rho\sigma}g^\mu{}_\nu \tag{2.15}$$

$$T^{\mu}{}_{\nu} = \frac{1}{2}F^{\rho\sigma}\big(\partial_\nu g_\rho{}^\mu A_\sigma - \partial_\nu A_\rho g_\sigma{}^\mu\big) - \frac{1}{4}F_{\rho\sigma}F^{\rho\sigma}g^\mu{}_\nu \tag{2.16}$$

$$T^{\mu}{}_{\nu} = F^{\mu\sigma}\partial_\nu A_\sigma - \frac{1}{4}F_{\rho\sigma}F^{\rho\sigma}g^\mu{}_\nu \tag{2.17}$$

which is the desired energy-momentum tensor.

As a sanity check, we can prove that it's indeed conserved when the equations of motion are valid,

$$\partial_\mu T^{\mu}{}_{\nu} = \partial_\mu F^{\mu\sigma}\partial_\nu A_\sigma + F^{\mu\sigma}\partial_\mu\partial_\nu A_\sigma - \frac{1}{2}F^{\rho\sigma}\partial_\mu F_{\rho\sigma}g^\mu{}_\nu \tag{2.18}$$

$$\partial_\mu T^{\mu}{}_{\nu} = F^{\mu\sigma}\partial_\mu\partial_\nu A_\sigma - F^{\rho\sigma}\partial_\nu\partial_\rho A_\sigma = 0 \tag{2.19}$$

$$\partial_\mu T^{\mu}{}_{\nu} = F^{\rho\sigma}(\partial_\rho\partial_\nu A_\sigma - \partial_\nu\partial_\rho A_\sigma) = 0 \tag{2.20}$$

showing that it's indeed conserved.

Problem 3

This problem is mainly another application of Noether's theorem. But, it serves as a nice example of a symmetry transformation which changes the space-time coordinates with an non unit determinant. This scale symmetry is a particular case of the more generic conformal transformations which is a whole area of research in QFT, it's also a nice exemple of an anomalous symmetry. Despite it being a symmetry of the classical theory, in most of the cases the scale symmetry is absent in the quantum theory, we probably will see this by the end of the course when we introduce beta functions and the anomalous mass dimension. There are a few known interacting theories which indeed retains scale/conformal symmetry after quantization.

Again we will use eq. (E.21), in which we just need to know the transformation law,

$$\begin{cases} x & \to x' = \mathrm{e}^a x \\ \phi(x) & \to \phi'(x') = \mathrm{e}^{-d_\phi a}\phi(x) \end{cases} \Rightarrow \left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}a}\right|_{a=0} = x^\mu, \quad \left.\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\right|_{a=0} = -d_\phi\phi(x) - \partial_\mu\phi(x)x^\mu \tag{3.1}$$

it's interesting to notice that the scale transformation actually "measures" the mass dimension of the objects being transformed, x has mass dimension $[x] = 1$, by consistency our field has mass dimension $[\phi] = -1 = -d_\phi$ which fixes the transformation law as $\phi'(x') \to \mathrm{e}^{a[\phi]}\phi(x)$. Also, is nice to check that $[\lambda] = 0$. We then start by showing that it's a symmetry,

$$S'_a[\phi'] = \int \mathrm{d}^4x' \left[-\frac{1}{2}\partial'_\mu\phi'(x')\partial'^\mu\phi'(x') - \frac{\lambda}{4!}\phi'^4(x')\right] \tag{3.2}$$

$$S'_a[\phi'] = \int \mathrm{d}^4x\, \mathrm{e}^{4a}\left[-\frac{1}{2}\mathrm{e}^{-2a-2d_\phi a}\partial_\mu\phi(x)\partial^\mu\phi(x) - \mathrm{e}^{-4d_\phi a}\frac{\lambda}{4!}\phi^4(x)\right] \tag{3.3}$$

$$S'_a[\phi'] = \int \mathrm{d}^4x \left[-\frac{1}{2}\mathrm{e}^{2a-2d_\phi a}\partial_\mu\phi(x)\partial^\mu\phi(x) - \mathrm{e}^{4a-4d_\phi a}\frac{\lambda}{4!}\phi^4(x)\right] \tag{3.4}$$

setting $d_\phi = 1$,

$$S'_a[\phi'] = \int \mathrm{d}^4x \left[-\frac{1}{2}\partial_\mu\phi(x)\partial^\mu\phi(x) - \frac{\lambda}{4!}\phi^4(x)\right] = S[\phi] \tag{3.5}$$

what settles down as a symmetry.

Now, let's compute the conserved charge, again, we use the above derived infinitesimal transformations in eq. (E.21),

$$D^\mu = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\left.\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\right|_{a=0} + \mathcal{L}\left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}a}\right|_{a=0} \tag{3.6}$$

$$D^\mu = -\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}(d_\phi\phi(x) + \partial_\alpha\phi(x)x^\alpha) + \mathcal{L}x^\mu \tag{3.7}$$

$$D^\mu = -d_\phi\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\phi(x) - \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\partial_\alpha\phi(x)x^\alpha + \mathcal{L}x^\mu \tag{3.8}$$

$$D^\mu = -d_\phi\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\phi(x) + x^\alpha\left(-\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\partial_\alpha\phi(x) + \mathcal{L}g^\mu{}_\alpha\right) \tag{3.9}$$

$$D^\mu = \partial^\mu\phi\phi + x^\alpha T^\mu{}_\alpha \tag{3.10}$$

in the last line we used $d_\phi = 1$ and identified the quantity multiplying x^α as the energy-momentum tensor. This format is more insightful, but for explicit calculations is better to expand into the explicit ϕ dependence,

$$D^\mu = \partial^\mu\phi(d_\phi\phi + \partial_\alpha\phi x^\alpha) + \left(-\frac{1}{2}\partial^\alpha\phi\partial_\alpha\phi - \frac{\lambda}{4!}\phi^4\right)x^\mu \tag{3.11}$$

Again, as a sanity check we show it is divergenceless,

$$\partial_\mu D^\mu = \partial_\mu\partial^\mu\phi(d_\phi\phi + \partial_\alpha\phi x^\alpha) + \partial^\mu\phi(d_\phi\partial_\mu\phi + \partial_\mu\partial_\alpha\phi x^\alpha + \partial_\alpha\phi\partial_\mu x^\alpha)$$
$$+ \left(-\partial^\alpha\phi\partial_\mu\partial_\alpha\phi - \frac{\lambda}{3!}\phi^3\partial_\mu\phi\right)x^\mu + \left(-\frac{1}{2}\partial^\alpha\phi\partial_\alpha\phi - \frac{\lambda}{4!}\phi^4\right)\partial_\mu x^\mu \tag{3.12}$$

$$\partial_\mu D^\mu = \Box\phi(d_\phi\phi + \partial_\alpha\phi x^\alpha) + \partial^\mu\phi\big(d_\phi\partial_\mu\phi + \partial_\mu\partial_\alpha\phi x^\alpha + \partial_\alpha\phi g_\mu{}^\alpha\big)$$
$$+ \left(-\partial^\alpha\phi\partial_\mu\partial_\alpha\phi - \frac{\lambda}{3!}\phi^3\partial_\mu\phi\right)x^\mu + 4\left(-\frac{1}{2}\partial^\alpha\phi\partial_\alpha\phi - \frac{\lambda}{4!}\phi^4\right) \tag{3.13}$$

$$\partial_\mu D^\mu = \left(\Box\phi - \frac{\lambda}{3!}\phi^3\right)x^\alpha\partial_\alpha\phi + \left(d_\phi\Box\phi - \frac{\lambda}{3!}\phi^3\right)\phi + \partial^\mu\phi(d_\phi\partial_\mu\phi + \partial_\mu\partial_\alpha\phi x^\alpha + \partial_\mu\phi)$$
$$- \partial^\alpha\phi\partial_\mu\partial_\alpha\phi x^\mu - 2\partial^\alpha\phi\partial_\alpha\phi \tag{3.14}$$

$$\partial_\mu D^\mu = \left(\Box\phi - \frac{\lambda}{3!}\phi^3\right)x^\alpha\partial_\alpha\phi + \left(\Box\phi - \frac{\lambda}{3!}\phi^3\right)\phi + (d_\phi - 1)\phi\Box\phi + (d_\phi - 1)\partial^\mu\phi\partial_\mu\phi \tag{3.15}$$

using the equation of motion,

$$\Box\phi = \frac{\lambda}{3!}\phi^3 \tag{3.16}$$

we get,

$$\partial_\mu D^\mu = (d_\phi - 1)\phi\Box\phi + (d_\phi - 1)\partial^\mu\phi\partial_\mu\phi \tag{3.17}$$

finally setting $d_\phi = 1$,

$$\partial_\mu D^\mu = 0 \tag{3.18}$$

Just as a curiosity, eq. (3.10) can be rewritten as,

$$D^\mu = \partial^\mu\left(\frac{1}{2}\phi^2\right) + x^\alpha T^\mu{}_\alpha \tag{3.19}$$

the first term is a total derivative, and usually total derivative contributions to the current can be ignored. For general theories the scale current has the form

$$D^\mu \sim x^\alpha T^\mu{}_\alpha + \text{ total derivative} \tag{3.20}$$

thus, it's conservation is dependent of, assuming translational invariance,

$$\partial_\mu D^\mu \sim \partial_\mu x^\alpha T^\mu{}_\alpha + x^\alpha\partial_\mu T^\mu{}_\alpha \tag{3.21}$$

$$\partial_\mu D^\mu \sim T^\mu{}_\mu \stackrel{?}{=} 0 \tag{3.22}$$

as long as the trace of the energy-momentum tensor is zero, up to total derivatives, the theory under consideration is scale invariant. This is all classical, under quantization the matter is way more intricate. In four dimensions and under the usual assumptions of unitarity, Poincaré invariance, locality, causality, all known interacting scale invariant theories are conformal ones. Example of non-interacting scale invariant theories are easy, any massless theory, in fact for pure electromagnetism we have $T^\mu{}_\mu = 0$. For interacting theories, QCD, or any Yang-Mills are classical scale invariant, but, under quantization aren't. This means the quantum energy momentum tensor has non-zero trace, despite the classical one being zero. Due to dimensional reasons the non-zero contribution generated by quantization comes accompanied by positive powers of $\hbar$,

$$\hat{T}^\mu{}_\mu = \hbar^n C \tag{3.23}$$

this phenomenon is usually called *Trace anomaly*. It plays an important role in String Theory also. Just to mention, one of the known interacting quantum conformally invariant theories is a so called $\mathcal{N} = 4$ Super-Yang-Mills.

Problem 4

The first two itens are just some well known properties of Lorentz transformations, the last item is other important application of Noether's theorem, which gives the angular momentum density tensor, an analogous quantity to the energy-momentum tensor, but for Lorentz transformations instead of translations.

4.A)

A Lorentz transformation has the following infinitesimal form when acting on a vector,

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu} = x^{\mu} + \omega^{\mu}{}_{\nu} x^{\nu} + \mathcal{O}\big(\omega^2\big) \tag{4.1}$$

it's defining property is to preserve the scalar product between any vectors,

$$\begin{aligned}
y^{\mu} x_{\mu} &= y'^{\mu} x'_{\mu} && (4.2)\\
y^{\mu} x_{\mu} &= \big(y^{\mu} + \omega^{\mu}{}_{\nu} y^{\nu} + \mathcal{O}\big(\omega^2\big)\big)\big(x_{\mu} + \omega_{\mu\alpha} x^{\alpha} + \mathcal{O}\big(\omega^2\big)\big) && (4.3)\\
y^{\mu} x_{\mu} &= y^{\mu} x_{\mu} + \omega^{\mu}{}_{\nu} y^{\nu} x_{\mu} + y^{\mu} \omega_{\mu\alpha} x^{\alpha} + \mathcal{O}\big(\omega^2\big) && (4.4)\\
0 &= \omega^{\mu}{}_{\nu} y^{\nu} x_{\mu} + y^{\mu} \omega_{\mu\alpha} x^{\alpha} + \mathcal{O}\big(\omega^2\big) && (4.5)
\end{aligned}$$

as ω is arbitrary, the only possible way the right-hand side is zero is if each coefficient of each order in ω is zero, at first order,

$$\begin{aligned}
0 &= \omega_{\mu\nu} y^{\nu} x^{\mu} + \omega_{\mu\nu} x^{\nu} y^{\mu} && (4.6)\\
0 &= y^{\nu} x^{\mu} \big(\omega_{\mu\nu} + \omega_{\nu\mu}\big) \Rightarrow \omega_{\mu\nu} + \omega_{\nu\mu} = 0 \Rightarrow \omega_{\mu\nu} = -\omega_{\nu\mu} && (4.7)
\end{aligned}$$

the lasts equalities follow from the equation being needed to hold for arbitrary values of x, y. We conclude that ω is anti-symmetric.

4.B)

A rotation should preserve the time component,

$$\begin{aligned}
x^0 = x'^0 &= x^0 + \omega^0{}_{\nu} x^{\nu} && (4.8)\\
0 &= \omega^0{}_{\nu} x^{\nu} \Rightarrow \omega^0{}_{\nu} = 0 \Rightarrow \omega_{0\nu} = 0 && (4.9)
\end{aligned}$$

together with the anti-symmetry gives,

$$\omega_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \omega_{12} & -\omega_{31} \\ 0 & -\omega_{12} & 0 & \omega_{23} \\ 0 & \omega_{31} & -\omega_{23} & 0 \end{pmatrix} \tag{4.10}$$

now, to understand better what each of these components are we can look at a general representation of a spatial rotation. Let $\boldsymbol{\theta}$ be the rotation direction and θ be it's modulus,

$$\mathbf{x}' = \mathbf{x}\cos\theta + \boldsymbol{\theta} \times \mathbf{x}\sin\theta + (\boldsymbol{\theta} \cdot \mathbf{x})\boldsymbol{\theta}(1 - \cos\theta) \tag{4.11}$$

to first order we have,

$$x'^i = \epsilon^i{}_{jk} \boldsymbol{\theta}^j x^k \theta = \omega^i{}_k x^k \Rightarrow \omega_{ik} = -\epsilon_{kij} \theta \boldsymbol{\theta}^j \tag{4.12}$$

we recognize from this last expression that $\theta\boldsymbol{\theta}^j$ is just the infinitesimal angle rotation in the $k - i$ plane, hence, we call it θ^j. Also, the equivalence between the two formulation of the

rotations implies that ω_{ik} is minus the angle of rotation in the $k-i$ plane. With this we can rewrite our matrix as,

$$\omega_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\theta^3 & \theta^2 \\ 0 & \theta^3 & 0 & -\theta^1 \\ 0 & -\theta^2 & \theta^1 & 0 \end{pmatrix} \tag{4.13}$$

4.C)

The first step to derive the Noether current is to state the Lorentz transformation for a scalar field,

$$\begin{cases} x^\mu & \to x'^\mu = \Lambda^\mu{}_\nu(\omega) x^\nu \\ \phi(x) & \to \phi'(x') = \phi(x) \end{cases} \tag{4.14}$$

which is parametrized by $\omega_{\mu\nu}$, now we should take the variation of the coordinates,

$$\frac{\mathrm{d}x'^\mu}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega_{\alpha\beta}=0} = \frac{\mathrm{d}}{\mathrm{d}\omega_{\alpha\beta}}\big(x^\mu + \omega^\mu{}_\nu x^\nu + \mathcal{O}(\omega^2)\big)\bigg|_{\omega_{\alpha\beta}=0} \tag{4.15}$$

may look trivial that $\frac{\mathrm{d}\omega_{\mu\nu}}{\mathrm{d}\omega_{\alpha\beta}} \stackrel{?}{=} g_\mu{}^\alpha g_\nu{}^\beta$, but this is wrong, as ω is anti-symmetric, it's true that $1 = \frac{\mathrm{d}\omega_{12}}{\mathrm{d}\omega_{12}} = -\frac{\mathrm{d}\omega_{21}}{\mathrm{d}\omega_{12}}$, the last expression doesn't takes into consideration this, it's necessary to antisymmetrize it, hence,

$$\frac{\mathrm{d}x'^\mu}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega_{\alpha\beta}=0} = \big(g_\alpha{}^\mu g_{\beta\nu} - g_\beta{}^\mu g_{\alpha\nu}\big)x^\nu = g_\alpha{}^\mu x_\beta - g_\beta{}^\mu x_\alpha \tag{4.16}$$

we also need the expression of the field variation, this is obtained by usual means of writing the transformation law as $\phi'(x) = \phi(\Lambda^{\mathrm{T}} x)$,

$$\frac{\mathrm{d}\phi'(x)}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} = \frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(\phi(\Lambda^{\mathrm{T}} x)\big)\bigg|_{\omega^{\alpha\beta}=0} = \partial_\nu\phi(x)\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(\Lambda^{\mathrm{T}} x\big)^\nu\bigg|_{\omega^{\alpha\beta}=0} \tag{4.17}$$

$$\frac{\mathrm{d}\phi'(x)}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} = \partial_\nu\phi(x)\frac{\mathrm{d}\Lambda_\rho{}^\nu}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} x^\rho = \partial_\nu\phi(x)\big(g_{\alpha\rho}g_\beta{}^\nu - g_{\beta\rho}g_\alpha{}^\nu\big)x^\rho \tag{4.18}$$

$$\frac{\mathrm{d}\phi'(x)}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} = x_\alpha\partial_\beta\phi - x_\beta\partial_\alpha\phi \tag{4.19}$$

again we use these variation expressions on eq. (E.21),

$$\tilde{M}^\mu{}_{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} + \mathcal{L}\frac{\mathrm{d}x'^\mu}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} \tag{4.20}$$

$$\tilde{M}^\mu{}_{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}(x_\alpha\partial_\beta\phi - x_\beta\partial_\alpha\phi) + \mathcal{L}\big(g_\alpha{}^\mu x_\beta - g_\beta{}^\mu x_\alpha\big) \tag{4.21}$$

$$\tilde{M}^\mu{}_{\alpha\beta} = -x_\alpha\left(-\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\partial_\beta\phi + \mathcal{L}g_\beta{}^\mu\right) + x_\beta\left(-\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\partial_\alpha\phi + \mathcal{L}g_\alpha{}^\mu\right) \tag{4.22}$$

$$\tilde{M}^\mu{}_{\alpha\beta} = x_\beta T^\mu{}_\alpha - x_\alpha T^\mu{}_\beta \tag{4.23}$$

where in the last line we identified the energy-momentum tensor. The conserved current is defined up to a constant, then we can change the sign of the quantity to our desire. It's more common to work with the opposite sign conserved current $M = -\tilde{M}$,

$$M^\mu{}_{\alpha\beta} = x_\alpha T^\mu{}_\beta - x_\beta T^\mu{}_\alpha \tag{4.24}$$

this sign choice is related to the conserved charge associated with M. Remember, the conserved charge given by the integral of the energy-momentum tensor is the four momentum,

$$P_\beta = \int \mathrm{d}^3\mathbf{x}\, T^0{}_\beta \tag{4.25}$$

thus, $T^0{}_\beta$ is the density of four momentum. With this information, the conserved quantity,

$$M_{\alpha\beta} = \int \mathrm{d}^3\mathbf{x}\, M^0{}_{\alpha\beta} = \int \mathrm{d}^3\mathbf{x}\, \left(x_\alpha T^0{}_\beta - x_\beta T^0{}_\alpha\right) \tag{4.26}$$

is the integral of position times density of momentum, anti-symmetrized. It's clear the resemblance to: $\mathbf{L} = \mathbf{x} \times \mathbf{p}$. We can even verify that $M_{\alpha\beta}$ as given by the above expression is a representation of the Lorentz algebra generators, but this representation lives in the field phase-space. That is, the algebra is obtained by the Poisson bracket,

$$\left\{M^{\alpha\beta}, M^{\mu\nu}\right\} = g^{\alpha\nu} M^{\mu\beta} - M^{\alpha\nu} g^{\beta\mu} - g^{\beta\nu} M^{\mu\alpha} + M^{\beta\nu} g^{\alpha\mu} \tag{4.27}$$

of course, without the i. This is so due to usually in classical mechanics we use the real version of the generators, as complex numbers are not so welcome.

Without a proper lagrangian, or other information about the theory, we cannot explicitly check that it's conserved. But, the condition of conservation of this current imposes other constrains in the theory,

$$\partial_\mu M^\mu{}_{\alpha\beta} = 0 = \partial_\mu\left(x_\alpha T^\mu{}_\beta - x_\beta T^\mu{}_\alpha\right) \tag{4.28}$$

$$0 = \partial_\mu x_\alpha T^\mu{}_\beta + x_\alpha \partial_\mu T^\mu{}_\beta - \partial_\mu x_\beta T^\mu{}_\alpha - x_\beta \partial_\mu T^\mu{}_\alpha \tag{4.29}$$

$$0 = g_{\mu\alpha} T^\mu{}_\beta + x_\alpha \partial_\mu T^\mu{}_\beta - g_{\mu\beta} T^\mu{}_\alpha - x_\beta \partial_\mu T^\mu{}_\alpha \tag{4.30}$$

$$0 = T_{\alpha\beta} - T_{\beta\alpha} + x_\alpha \partial_\mu T^\mu{}_\beta - x_\beta \partial_\mu T^\mu{}_\alpha \tag{4.31}$$

notice, if we have a Lorentz invariant scalar theory, which is also translational invariant, then,

$$0 = T_{\alpha\beta} - T_{\beta\alpha} \tag{4.32}$$

the energy-momentum tensor is symmetric!

Problem 5

Following the lead of last problem, this is again an application of the Noether's theorem to Lorentz transformations. But now with a difference of the field transforming non-trivially,

$$\begin{cases} x^\mu & \to x'^\mu = \Lambda^\mu{}_\nu(\omega)x^\nu \\ A^\rho(x) & \to A'^\rho(x') = \Lambda^\rho{}_\sigma A^\sigma(x) \end{cases} \tag{5.1}$$

the variation of coordinates we had already made, eq. (4.16),

$$\left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega_{\alpha\beta}=0} = g_\alpha{}^\mu x_\beta - g_\beta{}^\mu x_\alpha \tag{5.2}$$

Leaving only the variation of the field, which now has two contributions, the usual one due to x dependence, and another due to the field index transformation, it's not difficult to obtain the result, we just start from $A'^\rho(x) = \Lambda^\rho{}_\sigma A^\sigma\big(\Lambda^{\mathrm{T}}x\big)$, and apply the chain rule,

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = \left.\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(\Lambda^\rho{}_\sigma A^\sigma\big(\Lambda^{-1}x\big)\big)\right|_{\omega^{\alpha\beta}=0} \tag{5.3}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = \left.\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}(\Lambda^\rho{}_\sigma)\right|_{\omega^{\alpha\beta}=0} A^\sigma(x) + g^\rho{}_\sigma \left.\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(A^\sigma\big(\Lambda^{-1}x\big)\big)\right|_{\omega^{\alpha\beta}=0} \tag{5.4}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = \big(g_\alpha{}^\rho g_{\beta\sigma} - g_\beta{}^\rho g_{\alpha\sigma}\big)A^\sigma(x) + \partial_\nu A^\rho(x)\left.\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(\Lambda^{-1}x\big)^\nu\right|_{\omega^{\alpha\beta}=0} \tag{5.5}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha + \partial_\nu A^\rho(x)\left.\frac{\mathrm{d}\Lambda_\gamma{}^\nu}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} x^\gamma \tag{5.6}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha + \partial_\nu A^\rho(x)\big(g_{\alpha\gamma}g_\beta{}^\nu - g_{\beta\gamma}g_\alpha{}^\nu\big)x^\gamma \tag{5.7}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha + x_\alpha\partial_\beta A^\rho(x) - x_\beta\partial_\alpha A^\rho(x) \tag{5.8}$$

using these results in eq. (E.21), and as in the last problem, we set up the current with an additional sign due to convenience,

$$-M^\mu{}_{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\rho(x)}\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} + \mathcal{L}\left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} \tag{5.9}$$

$$-M^\mu{}_{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\rho(x)}\big(g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha + x_\alpha\partial_\beta A^\rho - x_\beta\partial_\alpha A^\rho\big) + \mathcal{L}\big(g_\alpha{}^\mu x_\beta - g_\beta{}^\mu x_\alpha\big) \tag{5.10}$$

$$\begin{aligned} -M^\mu{}_{\alpha\beta} = {} & x_\beta\left(-\frac{\partial\mathcal{L}}{\partial\partial_\mu A^\rho(x)}\partial_\alpha A^\rho + \mathcal{L}g_\alpha{}^\mu\right) - x_\alpha\left(-\frac{\partial\mathcal{L}}{\partial\partial_\mu A^\rho(x)}\partial_\beta A^\rho + \mathcal{L}g_\beta{}^\mu\right) \\ & + \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\rho(x)}\big(g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha\big) \end{aligned} \tag{5.11}$$

$$-M^\mu{}_{\alpha\beta} = x_\beta T^\mu{}_\alpha - x_\alpha T^\mu{}_\beta + \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\alpha(x)}A_\beta - \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\beta(x)}A_\alpha \tag{5.12}$$

in the last line we identified the energy-momentum tensor. Notice the form of the tensor,

$$M^\mu{}_{\alpha\beta} = x_\alpha T^\mu{}_\beta - x_\beta T^\mu{}_\alpha + S^\mu{}_{\alpha\beta}, \quad S^\mu{}_{\alpha\beta} = F^\mu{}_\alpha A_\beta - F^\mu{}_\beta A_\alpha \tag{5.13}$$

this form of the result we reached is very insightful. Of course the $xT - xT$ piece is referring to orbital angular momentum, while the S piece is about intrinsic angular momentum. Despite

this, M is not gauge invariant, as well as T isn't. One type of procedure to obtain an gauge invariant energy-momentum tensor and angular momentum tensor is the Belinfante one, first, we notice that due to conservation,

$$\partial_\mu M^\mu{}_{\alpha\beta} = T_{\alpha\beta} - T_{\beta\alpha} + \partial_\mu S^\mu{}_{\alpha\beta} = 0 \tag{5.14}$$

$$\partial_\mu S^\mu{}_{\alpha\beta} = T_{\beta\alpha} - T_{\alpha\beta} \tag{5.15}$$

the divergence of S measures exactly the anti-symmetry of T. Notice that we can extract the symmetric part of T as,

$$\tilde{\Theta}_{\alpha\beta} = \frac{1}{2}(T_{\alpha\beta} + T_{\beta\alpha}) \tag{5.16}$$

$$\tilde{\Theta}_{\alpha\beta} = T_{\alpha\beta} - \frac{1}{2}(T_{\alpha\beta} - T_{\beta\alpha}) \tag{5.17}$$

$$\tilde{\Theta}_{\alpha\beta} = T_{\alpha\beta} + \frac{1}{2}\partial_\mu S^\mu{}_{\alpha\beta} \tag{5.18}$$

this new tensor is still divergenceless, as $\partial^\alpha \partial_\mu S^\mu{}_{\alpha\beta} = 0$. We only didn't guaranteed that it's gauge invariant. Actually it isn't, as can be seen from the transformation,

$$A_\mu \to A_\mu + \partial_\mu \lambda \Rightarrow T_{\alpha\beta} \to T_{\alpha\beta} + F_\alpha{}^\sigma \partial_\sigma \partial_\beta \lambda \tag{5.19}$$

$$S^\mu{}_{\alpha\beta} \to S^\mu{}_{\alpha\beta} + F^\mu{}_\alpha \partial_\beta \lambda - F^\mu{}_\beta \partial_\alpha \lambda \Rightarrow \partial_\mu S^\mu{}_{\alpha\beta} \to \partial_\mu S^\mu{}_{\alpha\beta} + F^\mu{}_\alpha \partial_\mu \partial_\beta \lambda - F^\mu{}_\beta \partial_\mu \partial_\alpha \lambda \tag{5.20}$$

$$\tilde{\Theta}_{\alpha\beta} \to \tilde{\Theta}_{\alpha\beta} + \frac{1}{2}\big(F_\alpha{}^\mu \partial_\beta + F_\beta{}^\mu \partial_\alpha\big)\partial_\mu \lambda \tag{5.21}$$

Despite seeming unproductive, closer inspection in this last equations reveals that the first term in the variation of ∂S cancels exactly half the contribution of T, and as it contains $F^\mu{}_\alpha$, if we add another ∂S term anti-symmetrized in $\alpha\mu$ will be sufficient to cancel the T contribution. Now, there are the remaining changes coming from ∂S, which can be further eliminated by summing ∂S symmetrized in $\beta\mu$, thus, we define the *Belinfante improved energy-momentum tensor* as,

$$\Theta_{\alpha\beta} = T_{\alpha\beta} + \frac{1}{2}\partial_\mu \big[S^\mu{}_{\alpha\beta} - S_\alpha{}^\mu{}_\beta + S_{\beta\alpha}{}^\mu\big] \tag{5.22}$$

it is conserved, $\partial^\alpha \Theta_{\alpha\beta} = 0$, and it's gauge invariant (check it!). Is important to stress that despite we constructed it by using explicit example of electromagnetism, this formula is true for any Poincaré invariant theory. If we compute explicitly for the electromagnetism we get,

$$\Theta_{\alpha\beta} = F_\alpha{}^\mu F_{\beta\mu} - \frac{1}{4} g_{\alpha\beta} F_{\mu\nu} F^{\mu\nu} \tag{5.23}$$

Of course, we could define whatever we wanted as an *improved energy-momentum tensor*, the catch behind the Belinfante procedure is that this construction is unique, and also agrees with the more general Hilbert's energy-momentum tensor. To see that the Belinfante improvement is a honest construction we must obtain the same conserved charges from both the unimproved and the improved tensor.

$$\tilde{P}_\beta = \int \mathrm{d}^3\mathbf{x}\, \Theta^0{}_\beta = \int \mathrm{d}^3\mathbf{x} \left(T^0{}_\beta + \frac{1}{2}\partial_\mu \big[S^{\mu 0}{}_\beta - S^{0\mu}{}_\beta + S_\beta{}^{0\mu}\big] \right) \tag{5.24}$$

$$\tilde{P}_\beta = \int \mathrm{d}^3\mathbf{x}\, T^0{}_\beta + \text{ boundary contributions} \tag{5.25}$$

$$\tilde{P}_\beta \sim P_\beta \tag{5.26}$$

for reasonable physical field configurations we can most of the time neglect the boundary contributions.

We could use this inspiration to construct a gauge invariant angular momentum tensor, we first start with the orbital angular momentum using the improved energy-momentum tensor,

$$\Sigma^{\mu}{}_{\alpha\beta} = x_\alpha \Theta^{\mu}{}_{\beta} - x_\beta \Theta^{\mu}{}_{\alpha} \tag{5.27}$$

$$\Sigma^{\mu}{}_{\alpha\beta} = x_\alpha \left(T^{\mu}{}_{\beta} + \frac{1}{2}\partial_\nu \left[S^{\nu\mu}{}_{\beta} - S^{\mu\nu}{}_{\beta} + S_\beta{}^{\mu\nu} \right] \right) - x_\beta \left(T^{\mu}{}_{\alpha} + \frac{1}{2}\partial_\nu [S^{\nu\mu}{}_{\alpha} - S^{\mu\nu}{}_{\alpha} + S_\alpha{}^{\mu\nu}] \right) \tag{5.28}$$

$$\Sigma^{\mu}{}_{\alpha\beta} = M^{\mu}{}_{\alpha\beta} - S^{\mu}{}_{\alpha\beta} + x_\alpha \frac{1}{2}\partial_\nu \left[S^{\nu\mu}{}_{\beta} - S^{\mu\nu}{}_{\beta} + S_\beta{}^{\mu\nu} \right] - x_\beta \frac{1}{2}\partial_\nu [S^{\nu\mu}{}_{\alpha} - S^{\mu\nu}{}_{\alpha} + S_\alpha{}^{\mu\nu}] \tag{5.29}$$

now, integrating by parts and neglecting boundary terms, visually cued by $\sim$,

$$\Sigma^{\mu}{}_{\alpha\beta} \sim M^{\mu}{}_{\alpha\beta} - S^{\mu}{}_{\alpha\beta} - \partial_\nu x_\alpha \frac{1}{2}\left[S^{\nu\mu}{}_{\beta} - S^{\mu\nu}{}_{\beta} + S_\beta{}^{\mu\nu} \right] + \partial_\nu x_\beta \frac{1}{2}[S^{\nu\mu}{}_{\alpha} - S^{\mu\nu}{}_{\alpha} + S_\alpha{}^{\mu\nu}] \tag{5.30}$$

$$\Sigma^{\mu}{}_{\alpha\beta} \sim M^{\mu}{}_{\alpha\beta} - S^{\mu}{}_{\alpha\beta} - g_{\nu\alpha} \frac{1}{2}\left[S^{\nu\mu}{}_{\beta} - S^{\mu\nu}{}_{\beta} + S_\beta{}^{\mu\nu} \right] + g_{\nu\beta} \frac{1}{2}[S^{\nu\mu}{}_{\alpha} - S^{\mu\nu}{}_{\alpha} + S_\alpha{}^{\mu\nu}] \tag{5.31}$$

$$\Sigma^{\mu}{}_{\alpha\beta} \sim M^{\mu}{}_{\alpha\beta} - S^{\mu}{}_{\alpha\beta} - \frac{1}{2}\left[S_\alpha{}^{\mu}{}_{\beta} - S^{\mu}{}_{\alpha\beta} + S_\beta{}^{\mu}{}_{\alpha} \right] + \frac{1}{2}\left[S_\beta{}^{\mu}{}_{\alpha} - S^{\mu}{}_{\beta\alpha} + S_\alpha{}^{\mu}{}_{\beta} \right] \tag{5.32}$$

$$\Sigma^{\mu}{}_{\alpha\beta} \sim M^{\mu}{}_{\alpha\beta} - S^{\mu}{}_{\alpha\beta} + S^{\mu}{}_{\beta\alpha} \tag{5.33}$$

$$\Sigma^{\mu}{}_{\alpha\beta} \sim M^{\mu}{}_{\alpha\beta} \tag{5.34}$$

astonishing! The improved angular momentum tensor is given only by the orbital contribution with the improved energy-momentum tensor. Of course, the spin is still encoded inside Θ, but the decomposition of Σ into orbital plus intrinsic cannot be done in a gauge invariant manner. To choose a polarization basis for the electromagnetic field is to chose a gauge.

A Conventions

A.A) Metric

There are two choices of metric,

$$g_- = \mathrm{diag}\big(- \quad + \quad + \quad +\big), \quad g_+ = \mathrm{diag}\big(+ \quad - \quad - \quad -\big) \tag{A.1}$$

I will always use g_-. This changes a lot of signs in a lot of places, as example,

$$\mathcal{L} = g_+^{\mu\nu}\frac{1}{2}\partial_\mu\phi\partial_\nu\phi - V[\phi] = -g_-^{\mu\nu}\frac{1}{2}\partial_\mu\phi\partial_\nu\phi - V[\phi] \tag{A.2}$$

Physics should be invariant under conventions, so they don't matter, as long as you know which conventions you are using and are consistent.

A.B) Levi-Civita

The Levi-Civita is the (unique) totally antisymmetric four-pseudo-tensor, $\epsilon_{\alpha\beta\mu\nu}$, there are two possible choices of normalization,

$$\epsilon^+_{0123} = 1, \quad \epsilon^-_{0123} = -1 \tag{A.3}$$

I will always use ϵ^+. We define also the spatial Levi-Civita by restricting the covariant one,

$$\epsilon_{ijk} \coloneqq \epsilon_{0ijk} \tag{A.4}$$

B Lorentz Group Digression

The Lorentz group is defined as the set of all 4x4 matrices which preserve the scalar products with respect to the relativistic metric, that is, the set of all $\Lambda : x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu$, such,

$$x^\mu g_{\mu\nu} y^\nu = x'^\alpha g_{\alpha\beta} y'^\beta \tag{B.1}$$
$$x^\mu g_{\mu\nu} y^\nu = \Lambda^\alpha{}_\mu x^\mu g_{\alpha\beta} y^\nu \Lambda^\beta{}_\nu \tag{B.2}$$
$$x^\mu g_{\mu\nu} y^\nu = x^\mu y^\nu \Lambda^\alpha{}_\mu g_{\alpha\beta} \Lambda^\beta{}_\nu \tag{B.3}$$

as this should be valid for all x, y,

$$g_{\mu\nu} = \Lambda^{\mathrm{T}}{}_\mu{}^\alpha g_{\alpha\beta} \Lambda^\beta{}_\nu \tag{B.4}$$
$$g = \Lambda^{\mathrm{T}} g \Lambda \tag{B.5}$$

in other words, the Lorentz group is the group of all orthogonal 4x4 matrices with respect to the non-positive definite metric g. For orthogonal nxn matrices with respect to the standard $\mathbb{R}^n$ metric, this group is denoted $O(n)$, as we're in a 1 negative and 3 positive eigenvalues metric, we denote by $O(1,3) \cong O(3,1)$. Apart from the topological structure carried by $O(1,3)$, we are mainly interested in the tangent space structure at the origin, at the identity element, also know as the Lie algebra, $\mathfrak{o}(1,3)$. This is the case because the generators of the Lie algebra are in a one to one correspondence with hermitian operators in the Hilbert space of the quantum theory which represents conserved charges, for the Lorentz group these are the angular momentum and boost operators. It's not hard to obtain the connected piece of $O(1,3)$, we have two conditions which splits our group into 4 connected pieces,

$$\mathrm{Det}[g] = \mathrm{Det}[\Lambda^{\mathrm{T}}]\mathrm{Det}[g]\mathrm{Det}[\Lambda] \tag{B.6}$$
$$-1 = -\mathrm{Det}[\Lambda^{\mathrm{T}}]\mathrm{Det}[\Lambda] \tag{B.7}$$
$$1 = (\mathrm{Det}[\Lambda])^2 \Rightarrow \mathrm{Det}[\Lambda] = \pm 1 \tag{B.8}$$

and also,

$$g_{00} = \Lambda^{\mathrm{T}}{}_0{}^\alpha g_{\alpha\beta} \Lambda^\beta{}_0 \tag{B.9}$$
$$-1 = \Lambda^{\mathrm{T}}{}_0{}^0 g_{00} \Lambda^0{}_0 + \Lambda^{\mathrm{T}}{}_0{}^i g_{ij} \Lambda^j{}_0 \tag{B.10}$$
$$\Lambda^0{}_0 \Lambda^0{}_0 = 1 + \sum_i \left\| \Lambda^i{}_0 \right\|^2 \geq 1 \tag{B.11}$$
$$\left(\Lambda^0{}_0\right)^2 \geq 1 \Rightarrow \Lambda^0{}_0 \begin{cases} \geq 1 \\ \leq -1 \end{cases} \tag{B.12}$$

The connected piece is the one satisfying $\mathrm{Det}[\Lambda] = +1$ and $\Lambda^0{}_0 \geq 1$, which is called the proper orthochronous Lorentz group $SO^+(1,3)$, proper (S) for $\mathrm{Det}[\Lambda] = +1$ and orthochronous $(+)$ for $\Lambda^0{}_0 \geq 1$. It's not hard to see that from the connected piece we can get all the other three pieces by compositions with the following matrices,

$$\mathcal{P}^\mu{}_\nu = \mathrm{diag}\begin{pmatrix} 1 & -1 & -1 & -1 \end{pmatrix}, \quad \mathcal{T}^\mu{}_\nu = \mathrm{diag}\begin{pmatrix} -1 & 1 & 1 & 1 \end{pmatrix} \tag{B.13}$$
$$[\mathcal{P}\mathcal{T}]^\mu{}_\nu = \mathrm{diag}\begin{pmatrix} -1 & -1 & -1 & -1 \end{pmatrix} \tag{B.14}$$

in fact they form a group by themselves, $\{\mathbb{1}, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$, called the Klein four group. With them we can reconstruct the full Lorentz group,

$$O(1,3) = \{\mathbb{1}, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\} \rtimes SO^+(1,3) \tag{B.15}$$

The quantum theory allows for projective representations of groups, this is due to states of the Hilbert space which differ by a phase represent the same physical configuration, hence, representations of groups can obey the group multiplication law up to a phase, a.k.a. projective representations. Constructing all the projective representations is similar to constructing all the proper representations of a cover of the group, for quantum mechanics the double cover is enough. It's known that the double cover of $SO^+(1,3)$ is $SL(2,\mathbb{C})$, the group of 2x2 matrices of unit determinant over the complex numbers, that is,

$$SO^+(1,3) \cong SL(2,\mathbb{C})/\mathbb{Z}_2 \tag{B.16}$$

the real quantum symmetry group is then $SL(2,\mathbb{C})$, due to it being the double cover, it contains proper spinorial representations, which are not present at $SO^+(1,3)$, at least as proper representations.

Let's construct this isomorphism, first, a disclaimer, we'll use a little bit of a posteriori knowledge, which could be obtained through the item 1.*b* without knowing of this construction. What we need is a way to represent Lorentz vectors over the 2x2 complex values matrices, in which a proper orthochronous Lorentz transformation is given through a unit determinant transformation. The correct choice, which we'll leave to explain why latter,

$$X = \begin{pmatrix} -x^0 + x^3 & x^1 - \mathrm{i}x^2 \\ x^1 + \mathrm{i}x^2 & -x^0 - x^3 \end{pmatrix} = -x^0\mathbb{1} + x^i\sigma_i = x^\mu\sigma_\mu, \quad \mathrm{Det}[X] = -x_\mu x^\mu \tag{B.17}$$

with σ_i being the usual Pauli matrices, we also defined $\sigma_0 = -\mathbb{1}$, we leave for now open as if this is an honest four vector, but in fact it transforms covariantly over $SL(2,\mathbb{C})$, and even more, it's invariant. As the scalar product is given by the determinant, Lorentz transformations are linear matrix multiplications which preserve the determinant,

$$X' = \begin{pmatrix} -x'^0 + x'^3 & x'^1 - \mathrm{i}x'^2 \\ x'^1 + \mathrm{i}x'^2 & -x'^0 - x'^3 \end{pmatrix} = \tilde{U}X\tilde{V}, \quad \mathrm{Det}[X'] = \mathrm{Det}[X] \tag{B.18}$$

also, the transformation needs to preserve the reality of each of the components, this can be assured by noting that $X^\dagger = X$, and then demanding that $X'^\dagger = X'$, which fixes $\tilde{V} = \tilde{U}^\dagger$. The preservation of the determinant gives the following constraint,

$$\mathrm{Det}[X] = \mathrm{Det}[X'] \tag{B.19}$$

$$\mathrm{Det}[X] = \mathrm{Det}[\tilde{U}]\mathrm{Det}[X]\mathrm{Det}[\tilde{U}^\dagger] \tag{B.20}$$

$$1 = \left\|\mathrm{Det}[\tilde{U}]\right\|^2 \Rightarrow \mathrm{Det}[\tilde{U}] = \exp(\mathrm{i}\alpha) \tag{B.21}$$

notice that $\tilde{U}$ and $U = \exp\left(-\mathrm{i}\frac{\alpha}{2}\right)\tilde{U}$ generates the same transformation on X, hence, we can consider only the determinant one matrices. What we concluded up to now is that at least the 2x2 unit determinant matrices over the complexes is surjective in the (proper orthochronous?) Lorentz group, let's verify it's the proper orthochronous, to see the condition ${\Lambda^0}_0 \geq 1$, take $X = -\mathbb{1} \leftrightarrow x = \mathrm{diag}\begin{pmatrix}1 & 0 & 0 & 0\end{pmatrix}$, then,

$$X' = \begin{pmatrix} -{\Lambda^0}_0 + {\Lambda^3}_0 & {\Lambda^1}_0 - \mathrm{i}{\Lambda^2}_0 \\ {\Lambda^1}_0 + \mathrm{i}{\Lambda^2}_0 & -{\Lambda^0}_0 - {\Lambda^3}_0 \end{pmatrix} \Rightarrow \mathrm{Tr}[X'] = -2{\Lambda^0}_0 \tag{B.22}$$

thus,

$${\Lambda^0}_0 = -\frac{1}{2}\mathrm{Tr}[X'] = -\frac{1}{2}\mathrm{Tr}\left[UXU^\dagger\right] = \frac{1}{2}\mathrm{Tr}\left[UU^\dagger\right] \tag{B.23}$$

as $\mathrm{Det}[U] = 1$, we have $\mathrm{Det}[UU^\dagger] = 1$, if λ_1, λ_2 are the eigenvalues of U, λ_1^*, λ_2^* are the eigenvalues of $U^\dagger$, hence, the eigenvalues of $UU^\dagger$ are $\|\lambda_1\|^2, \|\lambda_2\|^2 > 0$, the constraint $\mathrm{Det}[UU^\dagger] = 1 = \|\lambda_1\|^2\|\lambda_2\|^2$ imposes that $\mathrm{Tr}\big[UU^\dagger\big] = \|\lambda_1\|^2 + \|\lambda_1\|^2 \geq 2$, therefore,

$$\Lambda^0{}_0 = \frac{1}{2}\,\mathrm{Tr}\big[UU^\dagger\big] \geq 1 \tag{B.24}$$

this clears that $SL(2,\mathbb{C})$ is at least surjective in the orthochronous Lorentz group. Now, the proper part, it suffices to show that there isn't in this group a parity transformation. Suppose exists U_p such that,

$$X' = U_p X U_p'^\dagger = U_p\big(-x^0\mathbb{1} + x^i\sigma_i\big)U_p^\dagger = -x^0\mathbb{1} - x^i\sigma_i \tag{B.25}$$

which imposes,

$$U_p U_p^\dagger = \mathbb{1} \tag{B.26}$$

$$U_p \sigma_i U_p^\dagger = -\sigma_i \tag{B.27}$$

Now, $\sigma_1\sigma_2\sigma_3 = \mathrm{i}\mathbb{1}$ (check it!), thus, we can compute the following,

$$U_p\sigma_1 U_p^\dagger U_p \sigma_2 U_p^\dagger U_p \sigma_3 U_p^\dagger = (-)^3\sigma_1\sigma_2\sigma_3 \tag{B.28}$$

$$U_p\sigma_1\sigma_2\sigma_3 U_p^\dagger = -\mathrm{i}\mathbb{1} \tag{B.29}$$

$$U_p \mathrm{i}\mathbb{1} U_p^\dagger = -\mathrm{i}\mathbb{1} \tag{B.30}$$

$$U_p U_p^\dagger = -\mathbb{1} \tag{B.31}$$

contradiction! Thus the parity transformation with $\mathrm{Det}[\mathcal{P}] = -1$ isn't present in $SL(2,\mathbb{C})$, and as this is a connected continuous group, none of the transformations with $\mathrm{Det}[\Lambda] = -1$ are present. This concludes that $SL(2,\mathbb{C})$ is surjective over the proper orthochronous Lorentz group. But is still up to discussion whether this is an injective map, the answer is, it isn't, more than one transformation in $SL(2,\mathbb{C})$ correspond to the same $SO^+(1,3)$ transformation. This can be seen from the transformation law,

$$X' = UXU^\dagger = (-U)X(-U)^\dagger \tag{B.32}$$

as U is two dimensional, $\mathrm{Det}[-U] = 1$, and we conclude that both U and $-U$ generate the same proper orthochronous Lorentz transformation, formally this means that $SO^+(1,3)$ can be seen as an equivalence class over $SL(2,\mathbb{C})$,

$$SO^+(1,3) \cong \{U \in SL(2,\mathbb{C}) | U \sim -U\} \tag{B.33}$$

the action $U \to -U$ can be seen as a $\mathbb{Z}_2$ action over $SL(2,\mathbb{C})$, hence, this equivalence class is a quotient group,

$$SO^+(1,3) \cong SL(2,\mathbb{C})/\mathbb{Z}_2 \tag{B.34}$$

this also means that $\mathfrak{o}(1,3) \cong \mathfrak{so}(1,3) \cong \mathfrak{so}^+(1,3) \cong \mathfrak{sl}(2,\mathbb{C})$. The presence of $\mathbb{Z}_2$ instead of $\mathbb{Z}_n$ signals that $SL(2,\mathbb{C})$ is the double-cover, actually in this case it is the universal-cover. For general dimensions the double-cover of $SO^+(p,q)$ is defined as $\mathrm{Spin}^+(p,q)$,

$$SO^+(p,q) \cong \mathrm{Spin}^+(p,q)/\mathbb{Z}_2 \tag{B.35}$$

in a comically similar fashion, we have the following,

$$SO(p,q) \cong \mathrm{Spin}(p,q)/\mathbb{Z}_2, \quad O(p,q) \cong \mathrm{Pin}(p,q)/\mathbb{Z}_2 \tag{B.36}$$

these groups $\mathrm{Spin}^+(p,q), \mathrm{Spin}(p,q), \mathrm{Pin}(p,q)$ are more easily definable by means of Clifford algebras, which we'll see soon enough, a representation of the Clifford algebra $\mathrm{Cl}(p,q)$ is given by the Dirac gamma matrices. The interesting point is that $\mathrm{Spin}(1,3) \cong \mathrm{Spin}(3,1)$ contains the $\mathcal{PT}$ transformation, but not parity and time inversion separately, but the Pin group, which contains all $\mathcal{P}, \mathcal{T}, \mathcal{PT}$ has the incredible property of

$$\mathrm{Pin}(1,3) \ncong \mathrm{Pin}(3,1) \tag{B.37}$$

which is a consequence of a non isomorphism between the Clifford algebras,

$$\mathrm{Cl}(1,3) \ncong \mathrm{Cl}(3,1) \tag{B.38}$$

This is simply outstanding, which in principle means that in a quantum theory with fermions and in which all $\mathcal{P}, \mathcal{T}, \mathcal{PT}$ are true symmetries there could be noticeable experimental results that could decide in favor of $\mathrm{Pin}(3,1)$ over $\mathrm{Pin}(1,3)$.

C Determinants

D Functional Derivative

First let's clear what is a functional, a functional is a map from some space of functions to the real numbers,

$$I : \mathcal{F}(\Omega \subset \mathbb{R}^m; \mathbb{R}^n) \to \mathbb{R} \tag{D.1}$$

$$f \mapsto I[f] \tag{D.2}$$

The most common way a functional can show up is by integration,

$$I[f] = \int_\Omega \mathrm{d}^m x \, \mathcal{I}[f(x)] \tag{D.3}$$

in these cases we have to guarantee that the domain of the functional is the space of functions for which the integral is well defined and converges,

$$f \in \mathcal{F}(\Omega \subset \mathbb{R}^m; \mathbb{R}^n) \Rightarrow I[f] = \int_\Omega \mathrm{d}^m x \, \mathcal{I}[f(x)] < \infty \tag{D.4}$$

now, a functional derivative, as all derivatives do, measures the diference with respect to change, in this case the difference is the functional value difference with respect to a change in the function it's evaluated in. As is the case in multivariate calculus, we can define the derivative implicitly by doing a directional derivative. Pick $\psi \in \mathcal{F}(\Omega \subset \mathbb{R}^m; \mathbb{R}^n)$ and $\epsilon \in \mathbb{R}$,

$$\left.\frac{\mathrm{d}I[f + \epsilon\psi]}{\mathrm{d}\epsilon}\right|_{\epsilon=0} = \int_\Omega \mathrm{d}^m y \, \frac{\delta I[f]}{\delta f(y)} \psi(y) \tag{D.5}$$

E Noether's Theorem

Here we're going to derive the Noether theorem for a handful of cases, first of all, the validity of this theorem is to global continuous symmetries, which in our field theory context means: we start of with an action which depends upon a field, $S[\phi]$, it's supposed to encode whatever index the fields have, then, a symmetry transformation is given in terms of:

$$\begin{cases} x^\mu & \to x'^\mu = x'^\mu(x;a) \\ \phi(x) & \to \phi'(x') = \phi'(\phi(x), x; a) \end{cases} \Rightarrow S[\phi] = S'_a[\phi'] \tag{E.1}$$

Let's look further at the symmetry condition,

$$S'_a[\phi'] - S[\phi] = \int\limits_{\Omega'} \mathrm{d}^4x'\, \mathcal{L}'[\phi'(x'); x'] - \int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L}[\phi(x); x] \tag{E.2}$$

$$\begin{aligned} S'_a[\phi'] - S[\phi] = &\int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L}'[\phi'(x); x] - \int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L}[\phi(x); x] \\ &+ \int\limits_{\Omega'} \mathrm{d}^4x'\, \mathcal{L}'[\phi'(x'); x'] - \int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L}'[\phi'(x); x] \end{aligned} \tag{E.3}$$

$$\begin{aligned} S'_a[\phi'] - S[\phi] = &\int\limits_{\Omega} \mathrm{d}^4x\, \{\mathcal{L}'[\phi'(x); x] - \mathcal{L}[\phi(x); x]\} \\ &+ \int\limits_{\Omega} \mathrm{d}^4x \left\{ \mathrm{Det}\left[\frac{\partial x'}{\partial x}\right] \mathcal{L}'[\phi'(x'); x'] - \mathcal{L}'[\phi'(x); x] \right\} \end{aligned} \tag{E.4}$$

$$\begin{aligned} S'_a[\phi'] - S[\phi] = &\int\limits_{\Omega} \mathrm{d}^4x\, \{\mathcal{L}'[\phi'(x); x] - \mathcal{L}[\phi(x); x]\} \\ &+ \int\limits_{\Omega} \mathrm{d}^4x \left\{ \mathrm{Det}\left[\frac{\partial x'}{\partial x}\right] - 1 \right\} \mathcal{L}'[\phi'(x'); x'] \\ &+ \int\limits_{\Omega} \mathrm{d}^4x\, \{\mathcal{L}'[\phi'(x'); x'] - \mathcal{L}'[\phi'(x); x]\} \end{aligned} \tag{E.5}$$

Suppose $\mathcal{L}' = \mathcal{L}$,

$$\begin{aligned} \left.\frac{[S'_a[\phi'] - S[\phi]]}{a}\right|_{a\to 0} = &\int\limits_{\Omega} \mathrm{d}^4x \int\limits_{\Omega} \mathrm{d}^4y \left\{ \frac{\delta\mathcal{L}}{\delta\phi(y)} \frac{\phi'(y) - \phi(y)}{a} \right\}\Bigg|_{a\to 0} \\ &+ \int\limits_{\Omega} \mathrm{d}^4x\, \frac{1}{a} \left\{ \mathrm{Det}\left[\frac{\partial x'}{\partial x}\right] - 1 \right\}\Bigg|_{a\to 0} \mathcal{L} \\ &+ \int\limits_{\Omega} \mathrm{d}^4x\, \frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha} \frac{x'^\alpha - x^\alpha}{a}\Bigg|_{a\to 0} \end{aligned}$$

$$\begin{aligned} \left.\frac{\mathrm{d}}{\mathrm{d}a} S'_a[\phi']\right|_{a=0} = &\int\limits_{\Omega} \mathrm{d}^4x \int\limits_{\Omega} \mathrm{d}^4y\, \frac{\delta\mathcal{L}}{\delta\phi(y)} \frac{\mathrm{d}\phi'(y)}{\mathrm{d}a}\Bigg|_{a=0} \\ &+ \int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L} \frac{\mathrm{d}}{\mathrm{d}a} \mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\Bigg|_{a=0} \end{aligned}$$

$$+\int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0} \tag{E.6}$$

Suppose the lagrangian has at most second derivatives,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_\Omega \mathrm{d}^4x \int_\Omega \mathrm{d}^4y \left[\frac{\partial\mathcal{L}}{\partial\phi(x)}\delta^{(4)}(x-y) + \frac{\partial\mathcal{L}}{\partial\partial_{x\mu}\phi(x)}\partial_{x\mu}\delta^{(4)}(x-y)\right]\frac{\mathrm{d}\phi'(y)}{\mathrm{d}a}\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}a}\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{E.7}$$

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_\Omega \mathrm{d}^4x \left[\frac{\partial\mathcal{L}}{\partial\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0} + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\partial_\mu\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0}\right] \\ &\quad + \int_\Omega \mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}a}\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{E.8}$$

Now impose the equations of motion,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_\Omega \mathrm{d}^4x \left[\partial_\mu\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0} + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\partial_\mu\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0}\right] \\ &\quad + \int_\Omega \mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}a}\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{E.9}$$

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_\Omega \mathrm{d}^4x\,\partial_\mu\left[\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0}\right] \\ &\quad + \int_\Omega \mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}a}\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{E.10}$$

The last missing piece is to compute the determinant,

$$\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right] = -\frac{1}{4!}\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\mu\nu\rho\sigma}\frac{\partial x'^\alpha}{\partial x^\mu}\frac{\partial x'^\beta}{\partial x^\nu}\frac{\partial x'^\gamma}{\partial x^\rho}\frac{\partial x'^\delta}{\partial x^\sigma} \tag{E.11}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = -\frac{1}{4!}\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\mu\nu\rho\sigma}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^\alpha}{\partial x^\mu}\frac{\partial x'^\beta}{\partial x^\nu}\frac{\partial x'^\gamma}{\partial x^\rho}\frac{\partial x'^\delta}{\partial x^\sigma}\right]\bigg|_{a=0} \tag{E.12}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = -\frac{1}{3!}\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\mu\nu\rho\sigma}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\right]\bigg|_{a=0}\frac{\partial x'^{\beta}}{\partial x^{\nu}}\bigg|_{a=0}\frac{\partial x'^{\gamma}}{\partial x^{\rho}}\bigg|_{a=0}\frac{\partial x'^{\delta}}{\partial x^{\sigma}}\bigg|_{a=0} \tag{E.13}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = -\frac{1}{3!}\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\mu\nu\rho\sigma}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\right]\bigg|_{a=0} g^{\beta}{}_{\nu}g^{\gamma}{}_{\rho}g^{\delta}{}_{\sigma} \tag{E.14}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = -\frac{1}{3!}\epsilon_{\alpha\nu\rho\sigma}\epsilon^{\mu\nu\rho\sigma}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\right]\bigg|_{a=0} \tag{E.15}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = \frac{1}{3!}3!g_{\alpha}{}^{\mu}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\right]\bigg|_{a=0} \tag{E.16}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = \frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\alpha}}\right]\bigg|_{a=0} = \frac{\mathrm{d}}{\mathrm{d}x^{\alpha}}\frac{\mathrm{d}x'^{\alpha}}{\mathrm{d}a}\bigg|_{a=0} \tag{E.17}$$

Thus,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_{\Omega}\mathrm{d}^4x\,\partial_{\mu}\left[\frac{\partial\mathcal{L}}{\partial\partial_{\mu}\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0}\right] \\ &\quad + \int_{\Omega}\mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}x^{\alpha}}\frac{\mathrm{d}x'^{\alpha}}{\mathrm{d}a}\bigg|_{a=0} \\ &\quad + \int_{\Omega}\mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^{\alpha}}\frac{\mathrm{d}x'^{\alpha}}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{E.18}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} = \int_{\Omega}\mathrm{d}^4x\,\frac{\mathrm{d}}{\mathrm{d}x^{\mu}}\left[\frac{\partial\mathcal{L}}{\partial\partial_{\mu}\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0} + \mathcal{L}\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}a}\bigg|_{a=0}\right] \tag{E.19}$$

It's trivial to include indicies,

$$\frac{\mathrm{d}}{\mathrm{d}a^i}S'_a[\phi'_I]\bigg|_{a^i=0} = \int_{\Omega}\mathrm{d}^4x\,\frac{\mathrm{d}}{\mathrm{d}x^{\mu}}\left[\frac{\partial\mathcal{L}}{\partial\partial_{\mu}\phi_I(x)}\frac{\mathrm{d}\phi'_I(x)}{\mathrm{d}a^i}\bigg|_{a^i=0} + \mathcal{L}\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}a^i}\bigg|_{a^i=0}\right] \tag{E.20}$$

So that the conserved current is,

$$J^{\mu}{}_{i} = \frac{\partial\mathcal{L}}{\partial\partial_{\mu}\phi_I(x)}\frac{\mathrm{d}\phi'_I(x)}{\mathrm{d}a^i}\bigg|_{a^i=0} + \mathcal{L}\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}a^i}\bigg|_{a^i=0} \tag{E.21}$$

F Noether's Second Theorem

This is an extension of the last result which deals with local transformations that leave the action invariant. Notice, we aren't saying symmetry, but invariance, this will be clear in the end of this appendix. Let the transformation be parametrized by a function $a(x)$,

$$\begin{cases} x^\mu & \to x'^\mu = x'^\mu(x; a(x)) \\ \phi(x) & \to \phi'(x') = \phi'(\phi(x), x; a(x)) \end{cases} \Rightarrow S[\phi] = S'[\phi'; a] \tag{F.1}$$

The reasoning proceeds as before, but now the derivatives are functional ones, also, due to distributional matters, is of need to include a test function,

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \frac{\delta \mathcal{L}}{\delta \phi(y)} \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \mathcal{L} \frac{\mathrm{d}}{\mathrm{d}x^\alpha} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \end{aligned} \tag{F.2}$$

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \left[\frac{\partial \mathcal{L}}{\partial \phi(x)} \delta^{(4)}(x-y) + \frac{\partial \mathcal{L}}{\partial \partial_{x\mu} \phi(x)} \partial_{x\mu} \delta^{(4)}(x-y) \right] \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\mathrm{d}}{\mathrm{d}x^\alpha} \left[\mathcal{L} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \right] \end{aligned} \tag{F.3}$$

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \frac{\partial \mathcal{L}}{\partial \phi(x)} \delta^{(4)}(x-y) \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \frac{\partial \mathcal{L}}{\partial \partial_{x\mu} \phi(x)} \partial_{x\mu} \delta^{(4)}(x-y) \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\mathrm{d}}{\mathrm{d}x^\alpha} \left[\mathcal{L} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \right] \end{aligned} \tag{F.4}$$

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\partial \mathcal{L}}{\partial \phi(x)} \frac{\delta \phi'(x)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \partial_{x\mu} \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \delta^{(4)}(x-y) \right] \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & - \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \partial_{x\mu} \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \delta^{(4)}(x-y) \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\mathrm{d}}{\mathrm{d}x^\alpha} \left[\mathcal{L} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \right] \end{aligned} \tag{F.5}$$

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, E[\phi(x); x] \frac{\delta \phi'(x)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \partial_{x\mu} \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \frac{\delta \phi'(x)}{\delta a(z)} \bigg|_{a \equiv 0} \right] \end{aligned}$$

$$+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}}{\mathrm{d}x^\alpha}\left[\mathcal{L}\frac{\delta x'^\alpha}{\delta a(z)}\bigg|_{a\equiv 0}\right] \tag{F.6}$$

Usually the space-time isn't transformed in a local transformation,

$$\begin{aligned}0 = &\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,E[\phi(x);x]\frac{\delta\phi'(x)}{\delta a(z)}\bigg|_{a\equiv 0}\\ &+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\,\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\delta\phi'(x)}{\delta a(z)}\bigg|_{a\equiv 0}\end{aligned} \tag{F.7}$$

As an example, let's use the electromagnetic case,

$$\begin{aligned}0 = &\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,E^\nu\frac{\delta A'_\nu(x)}{\delta a(z)}\bigg|_{a\equiv 0}\\ &+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\,\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu(x)}\frac{\delta A'_\nu(x)}{\delta a(z)}\bigg|_{a\equiv 0}\end{aligned} \tag{F.8}$$

$$\begin{aligned}0 = &\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,E^\nu\partial_{x\nu}\delta^{(4)}(x-z)\\ &+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\,\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu(x)}\partial_{x\nu}\delta^{(4)}(x-z)\end{aligned} \tag{F.9}$$

$$\begin{aligned}0 = -&\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,\partial_{x\nu}E^\nu\delta^{(4)}(x-z)\\ &+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\nu}\,E^\nu\delta^{(4)}(x-z)\\ &-\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\,\partial_{x\nu}\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu(x)}\delta^{(4)}(x-z)\end{aligned} \tag{F.10}$$

$$0 = -\int_\Omega \mathrm{d}^4x\,\partial_\mu E^\mu\psi(x) + \int_{\partial\Omega} \mathrm{d}^3S_\mu\left[E^\mu - \partial_\nu\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu(x)}\right]\psi(x) \tag{F.11}$$

This is all without imposing the equations of motion,

G Boundary Conditions

Let our spacetime be a four dimensional manifold Ω, with boundary $\partial\Omega = \Sigma_- \cup \Sigma_+ \cup \Gamma$. Our action principle is given by,

$$S[\phi] = \int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L}[\phi(x), x] + \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \ell^\alpha[\phi(x), x] \tag{G.1}$$

The variation of the action cannot have contributions from the bulk of Γ, and also is done with compact support outside of $\Sigma_- \cup \Sigma_+$, that is, on $\tilde{\Omega} = \Omega' \cup \Gamma'$,

$$\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\Omega} \mathrm{d}^4x\, \frac{\delta\mathcal{L}[x]}{\delta\phi(y)}\psi(y) + \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y) \tag{G.2}$$

$$\begin{aligned}\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = &\int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\Omega} \mathrm{d}^4x \left[\frac{\partial\mathcal{L}[x]}{\partial\phi(x)}\delta^{(4)}(x-y) + \frac{\partial\mathcal{L}[x]}{\partial\partial_\mu\phi(x)}\partial_\mu\delta^{(4)}(x-y)\right]\psi(y) \\ &+ \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y)\end{aligned} \tag{G.3}$$

$$\begin{aligned}\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = &\int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\Omega} \mathrm{d}^4x \left[\frac{\partial\mathcal{L}[x]}{\partial\phi(x)} - \partial_\mu\frac{\partial\mathcal{L}[x]}{\partial\partial_\mu\phi(x)}\right]\delta^{(4)}(x-y)\psi(y) \\ &+ \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\Omega} \mathrm{d}^4x\, \partial_{x\mu}\left[\frac{\partial\mathcal{L}[x]}{\partial\partial_\mu\phi(x)}\delta^{(4)}(x-y)\right]\psi(y) + \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y)\end{aligned} \tag{G.4}$$

$$\begin{aligned}\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = &\int\limits_{\tilde{\Omega}} \mathrm{d}^4y \left[\frac{\partial\mathcal{L}[y]}{\partial\phi(y)} - \partial_\mu\frac{\partial\mathcal{L}[y]}{\partial\partial_\mu\phi(y)}\right]\psi(y) \\ &+ \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\partial\mathcal{L}[x]}{\partial\partial_\alpha\phi(x)}\delta^{(4)}(x-y)\psi(y) + \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y)\end{aligned} \tag{G.5}$$

$$\begin{aligned}\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = &\int\limits_{\tilde{\Omega}} \mathrm{d}^4y \left[\frac{\partial\mathcal{L}[y]}{\partial\phi(y)} - \partial_\mu\frac{\partial\mathcal{L}[y]}{\partial\partial_\mu\phi(y)}\right]\psi(y) \\ &+ \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\partial\mathcal{L}[x]}{\partial\partial_\alpha\phi(x)}\psi(x) + \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y)\end{aligned} \tag{G.6}$$

For many cases $\ell^\alpha \equiv 0$ due to dynamical and covariance reasons. Of course ψ is zero on $\Sigma_\pm$, then, the equations of motion impose:

$$0 = \frac{\partial\mathcal{L}[\phi(x); x]}{\partial\phi(x)} - \partial_\mu\frac{\partial\mathcal{L}[\phi(x); x]}{\partial\partial_\mu\phi(x)}\bigg|_{x\in\Omega} \tag{G.7}$$

$$0 = n_\alpha(x)\partial^\alpha\phi(x)\bigg|_{x\in\Gamma} \tag{G.8}$$

With n_α being the normal direction in Γ. If we try to compute the conserved charge by Noether's theorem,

$$0 = \int\limits_{\partial\Omega} \mathrm{d}^3 S_\mu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \frac{\mathrm{d}\phi'(x)}{\mathrm{d}a} \bigg|_{a=0} + \mathcal{L} \frac{\mathrm{d}x'^\mu}{\mathrm{d}a} \bigg|_{a=0} \right] \tag{G.9}$$

$$0 = \int\limits_{\Sigma_+} \mathrm{d}^3 S_\mu \, J^\mu - \int\limits_{\Sigma_-} \mathrm{d}^3 S_\mu \, J^\mu + \int\limits_{\Gamma} \mathrm{d}^3 S_\mu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \frac{\mathrm{d}\phi'(x)}{\mathrm{d}a} \bigg|_{a=0} + \mathcal{L} \frac{\mathrm{d}x'^\mu}{\mathrm{d}a} \bigg|_{a=0} \right] \tag{G.10}$$

$$0 = \int\limits_{\Sigma_+} \mathrm{d}^3 S_\mu \, J^\mu - \int\limits_{\Sigma_-} \mathrm{d}^3 S_\mu \, J^\mu + \int\limits_{\Gamma} \mathrm{d}^3 S_\mu \, \mathcal{L} \frac{\mathrm{d}x'^\mu}{\mathrm{d}a} \bigg|_{a=0} \tag{G.11}$$

Now, if we have $\mathcal{L}\big|_{x\in\Gamma} = 0$ or $\frac{\mathrm{d}x'}{\mathrm{d}a}\big|_{a=0,x\in\Gamma} = 0$ then we get a conserved charge!

$$\int\limits_{\Sigma_+} \mathrm{d}^3 S_\mu \, J^\mu = \int\limits_{\Sigma_-} \mathrm{d}^3 S_\mu \, J^\mu \tag{G.12}$$

Now let's consider something other, let be a variation with change the values of the fields at the time-like boundaries, that is, a variation between different solutions,

$$\int\limits_{\Omega} \mathrm{d}^4 y \, \frac{\delta S[\phi]}{\delta \phi(y)} \psi_1(y) = \int\limits_{\Omega} \mathrm{d}^4 y \, E[\phi(y); y] \psi_1(y) + \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \, \Theta^\alpha[\phi(x); x] \psi_1(x) \tag{G.13}$$

$$\int\limits_{\Omega} \mathrm{d}^4 y \int\limits_{\Omega} \mathrm{d}^4 z \, \frac{\delta^2 S[\phi]}{\delta\phi(z)\delta\phi(y)} \psi_1(y)\psi_2(z) = \int\limits_{\Omega} \mathrm{d}^4 y \int\limits_{\Omega} \mathrm{d}^4 z \, \frac{\delta E[\phi(y); y]}{\delta\phi(z)} \psi_1(y)\psi_2(z) + \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \int\limits_{\Omega} \mathrm{d}^4 z \, \frac{\delta \Theta^\alpha[\phi(x); x]}{\delta\phi(z)} \psi_1(x)\psi_2(z) \tag{G.14}$$

$$\tag{G.15}$$

Antisymmetrize in $1 \leftrightarrow 2$,

$$0 = \int\limits_{\Omega} \mathrm{d}^4 y \int\limits_{\Omega} \mathrm{d}^4 z \left(\frac{\delta E[\phi(y); y]}{\delta\phi(z)} - \frac{\delta E[\phi(z); z]}{\delta\phi(y)} \right) \psi_1(y)\psi_2(z) + \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \int\limits_{\Omega} \mathrm{d}^4 z \left(\frac{\delta \Theta^\alpha[\phi(x); x]}{\delta\phi(z)} - \frac{\delta \Theta^\alpha[\phi(z); z]}{\delta\phi(x)} \right) \psi_1(x)\psi_2(z) \tag{G.16}$$

As long as the equation of motion have just even order of derivatives with constant coefficients,

$$0 = \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \int\limits_{\Omega} \mathrm{d}^4 z \left(\partial_x^\alpha \delta^{(4)}(x-z) - \partial_z^\alpha \delta^{(4)}(z-x) \right) \psi_1(x)\psi_2(z) \tag{G.17}$$

$$0 = \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \left(\psi_1(x) \partial^\alpha \psi_2(x) - \psi_2(x) \partial^\alpha \psi_1(x) \right) \tag{G.18}$$

Suppose that for a given solution of the equations of motion ϕ, it's true that $\phi+\psi_i$ are also solutions of the equations of motion, then,

$$\int\limits_{\Sigma_+} \mathrm{d}^3 S_\alpha \left(\psi_1(x)\partial^\alpha\psi_2(x) - \psi_2(x)\partial^\alpha\psi_1(x)\right) = \int\limits_{\Sigma_-} \mathrm{d}^3 S_\alpha \left(\psi_1(x)\partial^\alpha\psi_2(x) - \psi_2(x)\partial^\alpha\psi_1(x)\right) \tag{G.19}$$

In other words, the functional,

$$\Omega(\psi_1,\psi_2) = \int\limits_{\Sigma_t} \mathrm{d}^3\mathbf{x} \left(\psi_1\dot{\psi}_2 - \psi_2\dot{\psi}_1\right) \tag{G.20}$$

defined in the tangent space of the phase space of solutions, is independent of the time slice used to compute it. Taken into account that the phase space needs two variables for each degree of freedom, they are chosen as $\phi(\mathbf{x})$ and $\pi(\mathbf{x}) = \dot{\phi}(\mathbf{x})$, in this way, our functional can be rewritten as a phase space 2-form,

$$\Omega = \int\limits_{\Sigma_t} \mathrm{d}^3\mathbf{x}\, \delta\phi(\mathbf{x}) \wedge \delta\pi(\mathbf{x}) \tag{G.21}$$

The main point is that this 2-form is non degenerated and can be inverted as,

$$\{\cdot,\cdot\} = \int\limits_{\Sigma_t} \mathrm{d}^3\mathbf{x} \left(\frac{\delta}{\delta\phi(\mathbf{x})}\frac{\delta}{\delta\pi(\mathbf{x})} - \frac{\delta}{\delta\pi(\mathbf{x})}\frac{\delta}{\delta\phi(\mathbf{x})}\right) \tag{G.22}$$

Which provides the fundamental Poisson bracket,

$$\{\phi(t,\mathbf{x}),\pi(t,\mathbf{y})\} = \delta^{(3)}(\mathbf{x}-\mathbf{y}) \tag{G.23}$$

H Phase space structure of EM

Let our spacetime be a four dimensional manifold Ω, with boundary $\partial\Omega = \Sigma_- \cup \Sigma_+ \cup \Gamma$. Our action principle is given by,

$$S[A] = \int_\Omega \mathrm{d}^4x\, \mathcal{L}[\phi(x), x] = -\frac{1}{4}\int_\Omega \mathrm{d}^4x\, F^{\mu\nu}F_{\mu\nu} \tag{H.1}$$

The variation of the action cannot have contributions from the bulk of Γ, and also is done with compact support outside of $\Sigma_- \cup \Sigma_+$, that is, on $\tilde{\Omega} = \Omega' \cup \Gamma'$,

$$\int_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[A]}{\delta A_\alpha(y)}\psi_\alpha(y) = \int_{\tilde{\Omega}} \mathrm{d}^4y \int_\Omega \mathrm{d}^4x\, \frac{\delta \mathcal{L}[x]}{\delta A_\alpha(y)}\psi_\alpha(y) \tag{H.2}$$

$$\int_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[A]}{\delta A_\alpha(y)}\psi_\alpha(y) = \int_{\tilde{\Omega}} \mathrm{d}^4y \int_\Omega \mathrm{d}^4x \left[\frac{\partial \mathcal{L}[x]}{\partial A_\alpha(x)}\delta^{(4)}(x-y) + \frac{\partial \mathcal{L}[x]}{\partial\partial_\mu A_\alpha(x)}\partial_\mu\delta^{(4)}(x-y)\right]\psi_\alpha(y) \tag{H.3}$$

$$\begin{aligned}\int_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[A]}{\delta A_\alpha(y)}\psi_\alpha(y) = &\int_{\tilde{\Omega}} \mathrm{d}^4y \int_\Omega \mathrm{d}^4x \left[\frac{\partial \mathcal{L}[x]}{\partial A_\alpha(x)} - \partial_\mu\frac{\partial \mathcal{L}[x]}{\partial\partial_\mu A_\alpha(x)}\right]\delta^{(4)}(x-y)\psi_\alpha(y)\\ &+ \int_{\tilde{\Omega}} \mathrm{d}^4y \int_\Omega \mathrm{d}^4x\, \partial_{x\mu}\left[\frac{\partial \mathcal{L}[x]}{\partial\partial_\mu A_\alpha(x)}\delta^{(4)}(x-y)\right]\psi_\alpha(y)\end{aligned} \tag{H.4}$$

$$\begin{aligned}\int_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[A]}{\delta A_\alpha(y)}\psi_\alpha(y) = &\int_{\tilde{\Omega}} \mathrm{d}^4y \left[\frac{\partial \mathcal{L}[y]}{\partial A_\alpha(y)} - \partial_\mu\frac{\partial \mathcal{L}[y]}{\partial\partial_\mu A_\alpha(y)}\right]\psi_\alpha(y)\\ &+ \int_{\tilde{\Omega}} \mathrm{d}^4y \int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\, \frac{\partial \mathcal{L}[x]}{\partial\partial_\mu A_\alpha(x)}\delta^{(4)}(x-y)\psi_\alpha(y)\end{aligned} \tag{H.5}$$

$$\begin{aligned}\int_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[A]}{\delta A_\alpha(y)}\psi_\alpha(y) = &\int_{\tilde{\Omega}} \mathrm{d}^4y \left[\frac{\partial \mathcal{L}[y]}{\partial A_\alpha(y)} - \partial_\mu\frac{\partial \mathcal{L}[y]}{\partial\partial_\mu A_\alpha(y)}\right]\psi_\alpha(y)\\ &+ \int_{\partial\Omega} \mathrm{d}^3S_\mu\, \frac{\partial \mathcal{L}[x]}{\partial\partial_\mu A_\alpha(x)}\psi_\alpha(x)\end{aligned} \tag{H.6}$$

$$\begin{aligned}\int_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[A]}{\delta A_\alpha(y)}\psi_\alpha(y) = &\int_{\tilde{\Omega}} \mathrm{d}^4y\, E^\alpha[A(x); x]\psi_\alpha(y)\\ &+ \int_{\partial\Omega} \mathrm{d}^3S_\mu\, \Theta^{\mu\alpha}\psi_\alpha(x)\end{aligned} \tag{H.7}$$

With the identifications,

$$E^\alpha = \partial_\mu F^{\mu\alpha} \tag{H.8}$$

$$\Theta^{\mu\alpha} = -F^{\mu\alpha} \tag{H.9}$$

Again, we follow a similar reasoning,

$$\int_{\Omega} \mathrm{d}^4 y \, \frac{\delta S[A]}{\delta A_\alpha(y)} \psi_{1\alpha}(y) = \int_{\Omega} \mathrm{d}^4 y \, E^\alpha[y] \psi_{1\alpha}(y) + \int_{\partial\Omega} \mathrm{d}^3 S_\mu \, \Theta^{\mu\alpha}[x] \psi_{1\alpha}(x) \tag{H.10}$$

$$\int_{\Omega} \mathrm{d}^4 y \int_{\Omega} \mathrm{d}^4 z \, \frac{\delta^2 S[A]}{\delta A_\beta(z) \delta A_\alpha(y)} \psi_{1\alpha}(y) \psi_{2\beta}(z) = \int_{\Omega} \mathrm{d}^4 y \int_{\Omega} \mathrm{d}^4 z \, \frac{\delta E^\alpha[y]}{\delta A_\beta(z)} \psi_{1\alpha}(y) \psi_{2\beta}(z) + \int_{\partial\Omega} \mathrm{d}^3 S_\mu \int_{\Omega} \mathrm{d}^4 z \, \frac{\delta \Theta^{\mu\alpha}[x]}{\delta A_\beta(z)} \psi_{1\alpha}(x) \psi_{2\beta}(z) \tag{H.11}$$

$$\tag{H.12}$$

Antisymmetrize in $1 \leftrightarrow 2$,

$$0 = \int_{\Omega} \mathrm{d}^4 y \int_{\Omega} \mathrm{d}^4 z \left(\frac{\delta E^\alpha[y]}{\delta A_\beta(z)} - \frac{\delta E^\beta[z]}{\delta A_\alpha(y)} \right) \psi_{1\alpha}(y) \psi_{2\beta}(z) + \int_{\partial\Omega} \mathrm{d}^3 S_\mu \int_{\Omega} \mathrm{d}^4 z \left(\frac{\delta \Theta^{\mu\alpha}[x]}{\delta A_\beta(z)} - \frac{\delta \Theta^{\mu\beta}[z]}{\delta A_\alpha(x)} \right) \psi_{1\alpha}(x) \psi_{2\beta}(z) \tag{H.13}$$

As long as the equation of motion have just even order of derivatives with constant coefficients,

$$0 = - \int_{\partial\Omega} \mathrm{d}^3 S_\mu \int_{\Omega} \mathrm{d}^4 z \left(g^{\alpha\beta} \partial_x^\mu \delta^{(4)}(x-z) - g^{\mu\beta} \partial_x^\alpha \delta^{(4)}(x-z) - g^{\beta\alpha} \partial_z^\mu \delta^{(4)}(z-x) + g^{\mu\alpha} \partial_z^\beta \delta^{(4)}(z-x) \right) \psi_{1\alpha}(x) \psi_{2} \tag{H.14}$$

$$0 = - \int_{\partial\Omega} \mathrm{d}^3 S_\mu \left(\psi_{1\alpha} g^{\alpha\beta} \partial^\mu \psi_{2\beta} - \psi_{1\alpha} g^{\mu\beta} \partial^\alpha \psi_{2\beta} - \psi_{2\beta} g^{\beta\alpha} \partial^\mu \psi_{1\alpha} + \psi_{2\beta} g^{\mu\alpha} \partial^\beta \psi_{1\alpha} \right) \tag{H.15}$$

Suppose that for a given solution of the equations of motion A, it's true that $A + \psi_i$ are also solutions of the equations of motion, then,

$$\int_{\Sigma_+} \mathrm{d}^3 S_\mu \left(\psi_{1\alpha} (\partial^\mu \psi_2^\alpha - \partial^\alpha \psi_2^\mu) - \psi_{2\alpha} (\partial^\mu \psi_1^\alpha - \partial^\alpha \psi_1^\mu) \right) = \int_{\Sigma_-} \mathrm{d}^3 S_\mu \left(\psi_{1\alpha} (\partial^\mu \psi_2^\alpha - \partial^\alpha \psi_2^\mu) - \psi_{2\alpha} (\partial^\mu \psi_1^\alpha - \partial^\alpha \psi_1^\mu) \right) \tag{H.16}$$

In other words, the functional,

$$\Omega(\psi_1, \psi_2) = \int_{\Sigma_t} \mathrm{d}^3 \mathbf{x} \left(\psi_{1i} \left(\dot{\psi}_2^i - \partial^i \psi_{20} \right) - \psi_{2i} \left(\dot{\psi}_1^i - \partial^i \psi_{10} \right) \right) \tag{H.17}$$

defined in the tangent space of the phase space of solutions, is independent of the time slice used to compute it. The main problem here is that this functional is degenerated in the variables ψ_α, this means that they aren't a good parametrization of the phase space. To see that are degeneracies here,

$$\Omega(\psi_1 + \partial\lambda, \psi_2) = \Omega(\psi_1, \psi_2) + \int_{\Sigma_t} \mathrm{d}^3 \mathbf{x} \left(\partial_i \lambda \left(\dot{\psi}_2^i - \partial^i \psi_{20} \right) - \psi_{2i} \left(\partial^i \dot{\lambda} - \partial^i \dot{\lambda} \right) \right) \tag{H.18}$$

$$\Omega(\psi_1 + \partial\lambda, \psi_2) = \Omega(\psi_1, \psi_2) + \int\limits_{\partial\Sigma_t} \mathrm{d}^2 S_i \, \lambda\Big(\dot{\psi}_2^i - \partial^i \psi_{20}\Big) - \int\limits_{\Sigma_t} \mathrm{d}^3\mathbf{x} \, \lambda \partial_i \Big(\dot{\psi}_2^i - \partial^i \psi_{20}\Big) \tag{H.19}$$

$$\Omega(\psi_1 + \partial\lambda, \psi_2) = \Omega(\psi_1, \psi_2) + \int\limits_{\partial\Sigma_t} \mathrm{d}^2 S_i \, \lambda\Big(\dot{\psi}_2^i - \partial^i \psi_{20}\Big) \tag{H.20}$$

as long as λ is well behaved in the spatial boundary, the last integral is zero, and thus we have that the Ω is degenerated in this space, also, we can shift as we please the canonical momentum $\dot{\psi}_0$, with no impact on Ω. In total, we should have just two coordinates and conjugated momenta, the usual choice is,

Taken into account that the phase space needs two variables for each degree of freedom, they are chosen as $\phi(\mathbf{x})$ and $\pi(\mathbf{x}) = \dot{\phi}(\mathbf{x})$, in this way, our functional can be rewritten as a phase space 2-form,

$$\Omega = \int\limits_{\Sigma_t} \mathrm{d}^3\mathbf{x} \, \delta\phi(\mathbf{x}) \wedge \delta\pi(\mathbf{x}) \tag{H.21}$$

The main point is that this 2-form is non degenerated and can be inverted as,

$$\{\cdot, \cdot\} = \int\limits_{\Sigma_t} \mathrm{d}^3\mathbf{x} \left(\frac{\delta}{\delta\phi(\mathbf{x})} \frac{\delta}{\delta\pi(\mathbf{x})} - \frac{\delta}{\delta\pi(\mathbf{x})} \frac{\delta}{\delta\phi(\mathbf{x})} \right) \tag{H.22}$$

Which provides the fundamental Poisson bracket,

$$\{\phi(t, \mathbf{x}), \pi(t, \mathbf{y})\} = \delta^{(3)}(\mathbf{x} - \mathbf{y}) \tag{H.23}$$

I Algebra of conserved quantities

Given an internal symmetry of our action,

$$\begin{cases} \phi^I(x) & \to \phi'^I(x;a) \\ x^\mu & \to x'^\mu = x^\mu \end{cases} \Rightarrow S[\phi] = S'_a[\phi] \tag{I.1}$$

where I encode whatever index the fields of our theory may have. Suppose this symmetry is represented by some Lie Group/Algebra, that is, we have,

$$\phi'^I(x) = \exp\left(-\mathrm{i}a_i T^i\right)^I{}_J \phi^J(x) \Rightarrow \left.\frac{\mathrm{d}\phi'^I(x)}{\mathrm{d}a^i}\right|_{a^i=0} = -\mathrm{i}T^I{}_{Ji}\,\phi^J(x) \tag{I.2}$$

Consider the conserved current associated with this symmetry,

$$J^\mu{}_i = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi^I(x)}\left.\frac{\mathrm{d}\phi'^I(x)}{\mathrm{d}a^i}\right|_{a^i=0} \tag{I.3}$$

$$J^\mu{}_i = (-)^2\partial^\mu\phi_I(x)\mathrm{i}T^I{}_{Ji}\,\phi^J(x) \tag{I.4}$$

$$J^0{}_i = -\pi_I\mathrm{i}T^I{}_{Ji}\,\phi^J \tag{I.5}$$

Where we made use of the canonical conjugated momenta $\pi_I = \frac{\partial\mathcal{L}}{\partial\dot{\phi}^I}$. Now compute the conserved charge,

$$Q_i = \int \mathrm{d}^3\mathbf{x}\, J^0{}_i = -\int \mathrm{d}^3\mathbf{x}\,\pi_I\mathrm{i}T^I{}_{Ji}\,\phi^J \tag{I.6}$$

Remember from classical mechanics, in the Hamiltonian formalism we have as dynamical variables the generalized coordinate $\phi^I(t,\mathbf{x})$, and it's canonical conjugated momentum $\pi_I(t,\mathbf{x})$, they define the phase space and can be used to construct the Poisson bracket,

$$\{A,B\} = \int \mathrm{d}^3\mathbf{x}\left(\frac{\delta A}{\delta\phi^I(\mathbf{x})}\frac{\delta B}{\delta\pi_I(\mathbf{x})} - \frac{\delta A}{\delta\pi_I(\mathbf{x})}\frac{\delta B}{\delta\phi^I(\mathbf{x})}\right) \tag{I.7}$$

as this is a theory of the continuum, derivatives have to be exchanged by functional derivatives and sum over canonical pairs have to be exchanged for integration, still, it satisfy the canonical equal time Poisson bracket relations, $\left\{\phi^I(t,\mathbf{x}),\pi_J(t,\mathbf{y})\right\} = \delta^I{}_J\delta^{(3)}(\mathbf{x}-\mathbf{y})$ (verify!). It's interesting to compute the Poisson bracket between two conserved charges,

$$\{Q_i,Q_j\} = \int \mathrm{d}^3\mathbf{x}\left(\frac{\delta Q_i}{\delta\phi^I(\mathbf{x})}\frac{\delta Q_j}{\delta\pi_I(\mathbf{x})} - \frac{\delta Q_i}{\delta\pi_I(\mathbf{x})}\frac{\delta Q_j}{\delta\phi^I(\mathbf{x})}\right) \tag{I.8}$$

$$\{Q_i,Q_j\} = (-)^2\int \mathrm{d}^3\mathbf{x}\left(\pi_K\mathrm{i}T^K{}_{Ii}\,\mathrm{i}T^I{}_{Jj}\,\phi^J - \mathrm{i}T^I{}_{Ji}\,\phi^J\pi_K\mathrm{i}T^K{}_{Ij}\right) \tag{I.9}$$

$$\{Q_i,Q_j\} = \mathrm{i}^2\int \mathrm{d}^3\mathbf{x}\,\pi_K\left(T^K{}_{Ii}T^I{}_{Jj} - T^K{}_{Ij}T^I{}_{Ji}\right)\phi^J \tag{I.10}$$

$$\{Q_i,Q_j\} = \mathrm{i}^2\int \mathrm{d}^3\mathbf{x}\,\pi_K\Big([T_i,T_j]^K{}_J\Big)\phi^J \tag{I.11}$$

Now, as $T^K{}_{Ii}$ are the generators of a Lie Algebra, they obey an commutation relation of the form, $[T_i,T_j] = \mathrm{i}f_{ij}{}^kT_k$, thus,

$$\{Q_i,Q_j\} = \mathrm{i}^2\int \mathrm{d}^3\mathbf{x}\,\pi_K\left(\mathrm{i}f_{ij}{}^kT^K{}_{Jk}\right)\phi^J \tag{I.12}$$

$$\{Q_i, Q_j\} = -\mathrm{i}^2 f_{ij}{}^k (-) \int \mathrm{d}^3\mathbf{x}\, \pi_K \mathrm{i} T^K{}_{Jk}\, \phi^J \tag{I.13}$$

$$\{Q_i, Q_j\} = f_{ij}{}^k Q_k \tag{I.14}$$

The classical charges satisfy an algebra with the same structure constants as the symmetry group! In the quantum theory, at least from the canonical quantization POV, we have a correspondence between the commutator of two operators, and the Poisson bracket between the classical observables, $\left[\hat{A}, \hat{B}\right] \to \mathrm{i}\hbar\{A, B\}$, therefore, each classical conserved charge should be correspondent to a conserved charge operator $\hat{Q}_i$ which *should* satisfy the same Lie Algebra,

$$\left[\hat{Q}_i, \hat{Q}_j\right] = \mathrm{i}\hbar f_{ij}{}^k \hat{Q}_k \tag{I.15}$$

Almost always this isn't the case, and the symmetry has a anomaly in the quantum theory, that is, a transformation which is a symmetry in the classical theory isn't anymore a symmetry when the theory is quantized. For these cases, the most common and physical relevant way an anomaly shows is through a central extension of the Lie Algebra,

$$\left[\hat{Q}_i, \hat{Q}_j\right] = \mathrm{i}\hbar f_{ij}{}^k \hat{Q}_k + \hbar^2 C_{[ij]} \mathbb{1} \tag{I.16}$$

The name *central extension* is so because $C_{[ij]}$ are just numbers, that is, they commute with every other generator of the algebra — $\left[C_{[ij]}, \hat{Q}_k\right] = 0$ —, and thus, they belong to the center of the algebra — the maximum set of mutually commuting generators —. The $\hbar^2$ is needed for dimensional reasons (check it!), and it explain why this don't show up in the classical theory, if we consider the limit of,

$$\{Q_i, Q_f\} \coloneqq \lim_{\hbar \to 0} \frac{1}{\mathrm{i}\hbar} \left[\hat{Q}_i, \hat{Q}_j\right] = f_{ij}{}^k Q_k \tag{I.17}$$

That is, the $\hbar^2$ term drop off, it's a true quantum effect. There are other ways of understanding why the central extensions of algebras show up in the process of quantization, there is a close relation with the need to consider the double cover of Lie Groups/Algebras when going from classical symmetries to quantum symmetries, and also some overlap with the need of normal ordering quantum operators. Most of the anomalies are harmless, but, if there is an gauge anomaly the theory is physically inconsistent.

Luckily, the Poincaré Group/Algebra does not allow any kind of central extension (see Weinberg for proof), hence, the Poincaré symmetry is realized correctly in the quantum theory.

Other relevant property of the conserved charges is: they generate the symmetry transformation in the phase space variables,

$$\left\{Q_i, \phi^I(x)\right\} = \int \mathrm{d}^3\mathbf{y} \left(\frac{\delta Q_i}{\delta \phi^K(\mathbf{y})} \frac{\delta \phi^I(t, \mathbf{x})}{\delta \pi_K(\mathbf{y})} - \frac{\delta Q_i}{\delta \pi_K(\mathbf{y})} \frac{\delta \phi^I(t, \mathbf{x})}{\delta \phi^K(\mathbf{y})} \right) \tag{I.18}$$

$$\left\{Q_i, \phi^I(x)\right\} = (-)^2 \int \mathrm{d}^3\mathbf{y}\, \mathrm{i} T^K{}_{Ji}\, \phi^J \delta^I{}_K \delta^{(3)}(\mathbf{x} - \mathbf{y}) \tag{I.19}$$

$$\left\{Q_i, \phi^I(x)\right\} = \mathrm{i} T^I{}_{Ji}\, \phi^J \equiv \left.\frac{\mathrm{d}\phi'^I}{\mathrm{d}a^i}\right|_{a^I=0} \Rightarrow \left[\hat{Q}_i, \hat{\phi}^I(x)\right] = \mathrm{i}\hbar \left.\frac{\mathrm{d}\hat{\phi}'^I}{\mathrm{d}a^i}\right|_{a^I=0} \tag{I.20}$$

The same reasoning can be applied to space-time symmetries (try it!).